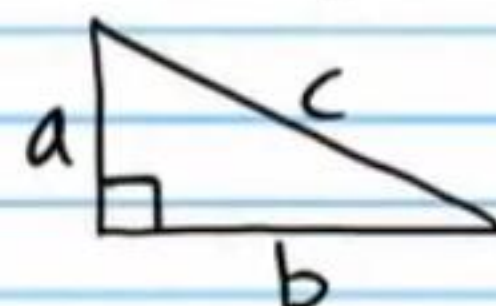


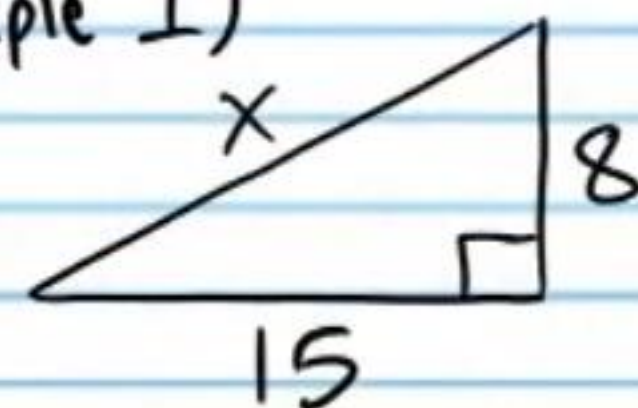
Using the Pythagorean Theorem to Find the Values of Trigonometric Functions

The Pythagorean Theorem: If a , b , and c are the legs and hypotenuse of a right triangle, respectively, then $a^2 + b^2 = c^2$.



Find the missing side lengths.

• Example I)



$$a^2 + b^2 = c^2$$

$$15^2 + 8^2 = c^2$$

$$225 + 64 = c^2$$

$$289 = c^2$$

$$\pm\sqrt{289} = \sqrt{c^2}$$

$$\pm 17 = c$$

$$\boxed{17 = c}$$

Example II)



$$a^2 + b^2 = c^2$$

$$3^2 + y^2 = (\sqrt{15})^2$$

$$9 + y^2 = 15$$

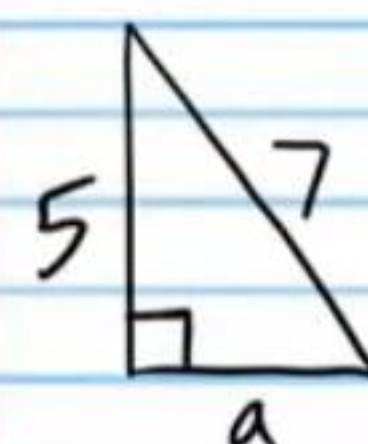
$$y^2 = 6$$

$$\sqrt{y^2} = \pm\sqrt{6}$$

$$y = \pm\sqrt{6}$$

$$\boxed{y = \sqrt{6}}$$

Example III)



$$a^2 + b^2 = c^2$$

$$a^2 + 5^2 = 7^2$$

$$a^2 + 25 = 49$$

$$a^2 = 24$$

$$\sqrt{a^2} = \pm\sqrt{24}$$

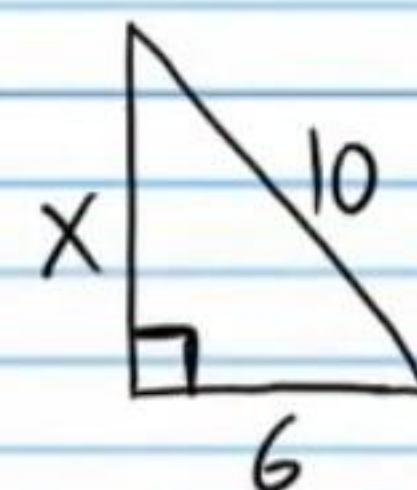
$$a = \pm\sqrt{4 \cdot 6}$$

$$a = \pm\sqrt{4} \cdot \sqrt{6}$$

$$a = \pm 2 \cdot \sqrt{6}$$

$$\boxed{a = 2\sqrt{6}}$$

• Example IV)



$$a^2 + b^2 = c^2$$

$$x^2 + 6^2 = 10^2$$

$$x^2 + 36 = 100$$

$$x^2 = 64$$

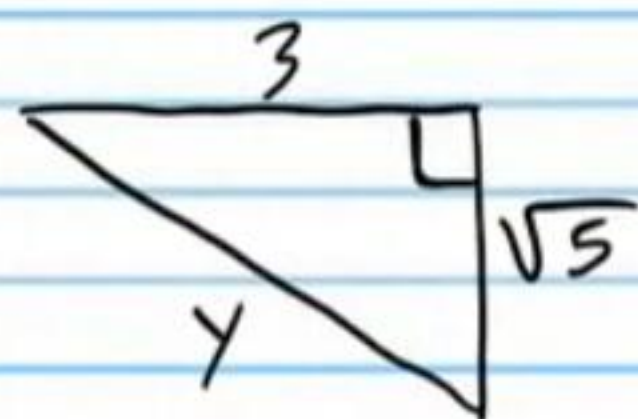
$$\sqrt{x^2} = \pm\sqrt{64}$$

$$x = \pm 8$$

$$\boxed{x = 8}$$

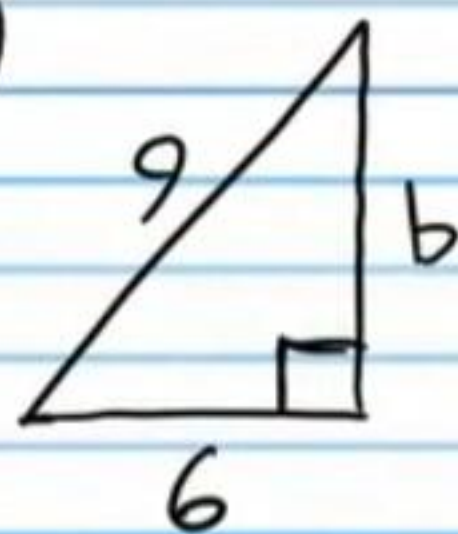
$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$

Example V)



$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 3^2 + (\sqrt{5})^2 &= c^2 \\
 9 + 5 &= c^2 \\
 14 &= c^2 \\
 \pm\sqrt{14} &= \sqrt{c^2} \\
 \pm\sqrt{14} &= c \\
 \boxed{c = \sqrt{14}}
 \end{aligned}$$

Example VI)



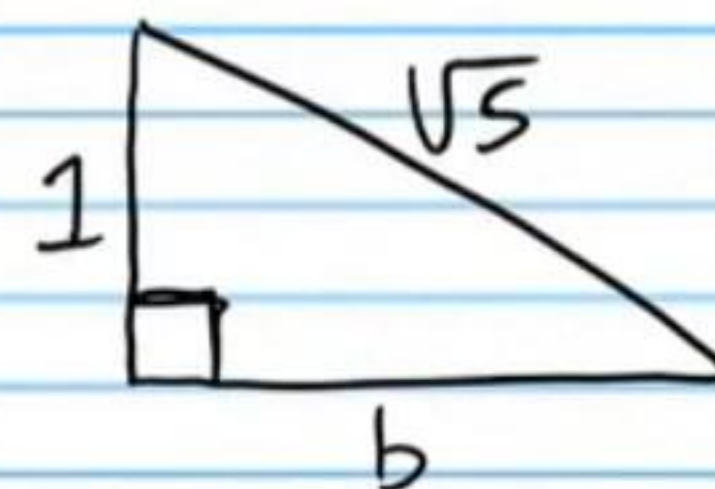
$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 6^2 + b^2 &= 9^2 \\
 36 + b^2 &= 81 \\
 b^2 &= 45 \\
 \sqrt{b^2} &= \pm\sqrt{45} \\
 \sqrt{a \cdot b} &= \sqrt{a} \cdot \sqrt{b} \quad b = \pm\sqrt{9 \cdot 5} \\
 b &= \pm\sqrt{9} \cdot \sqrt{5} \\
 b &= \pm 3\sqrt{5} \\
 \boxed{b = 3\sqrt{5}}
 \end{aligned}$$

Finding the Five Remaining Trigonometric Ratios Given One

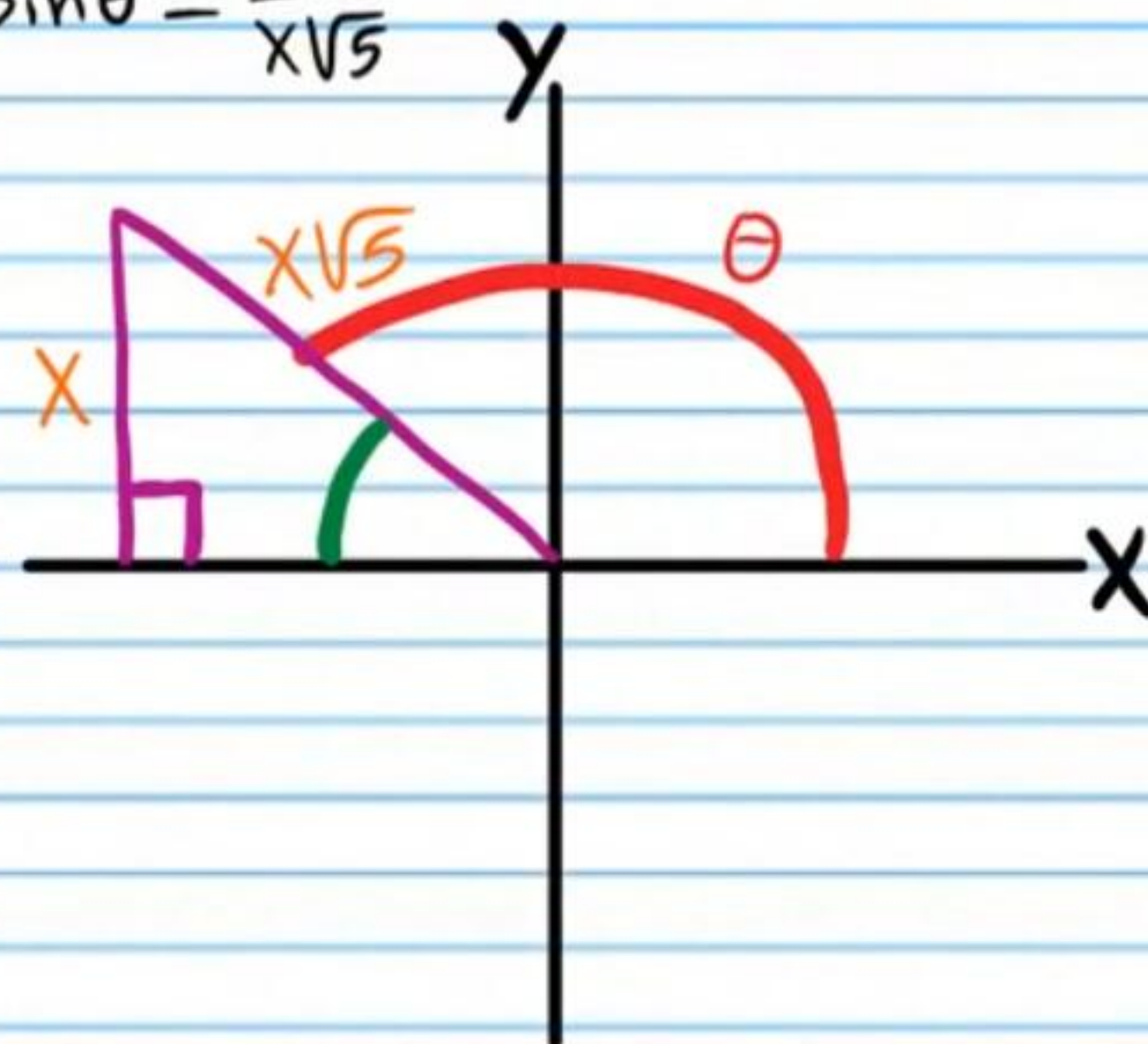
- ① If $\sin \theta = \frac{1}{\sqrt{5}}$, and θ terminates in quadrant II, find the values of the five remaining basic trigonometric functions of θ .

$$\sin \theta = \frac{x}{x\sqrt{5}}$$

Let $x=1$



$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 1^2 + b^2 &= (\sqrt{5})^2 \\
 1 + b^2 &= 5 \\
 b^2 &= 4 \\
 b &= \pm 2
 \end{aligned}$$



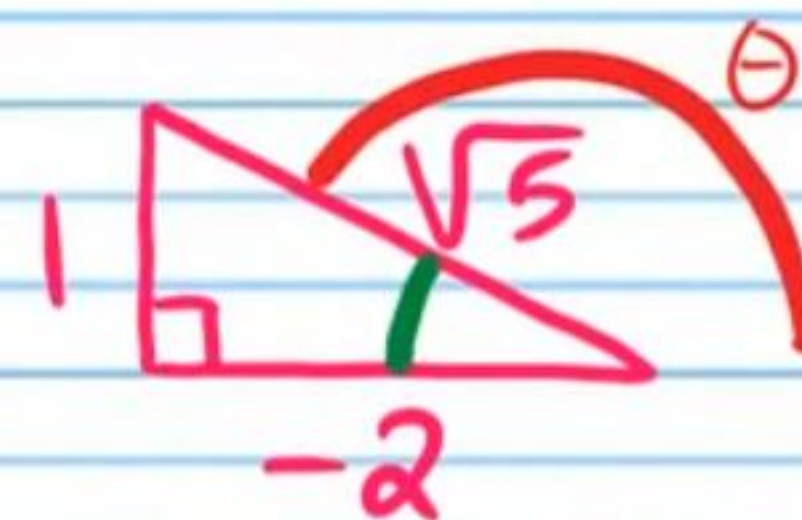
$$\text{i) } \csc \theta = \frac{\sqrt{5}}{1} = \boxed{\sqrt{5}}$$

$$\text{ii) } \sec \theta = \frac{\sqrt{5}}{-2} = \boxed{\frac{\sqrt{5}}{-2}}$$

$$\text{iii) } \cot \theta = \frac{-2}{1} = \boxed{-2}$$

$$\text{iv) } \cos \theta = \frac{-2}{\sqrt{5}} = \boxed{-\frac{2\sqrt{5}}{5}}$$

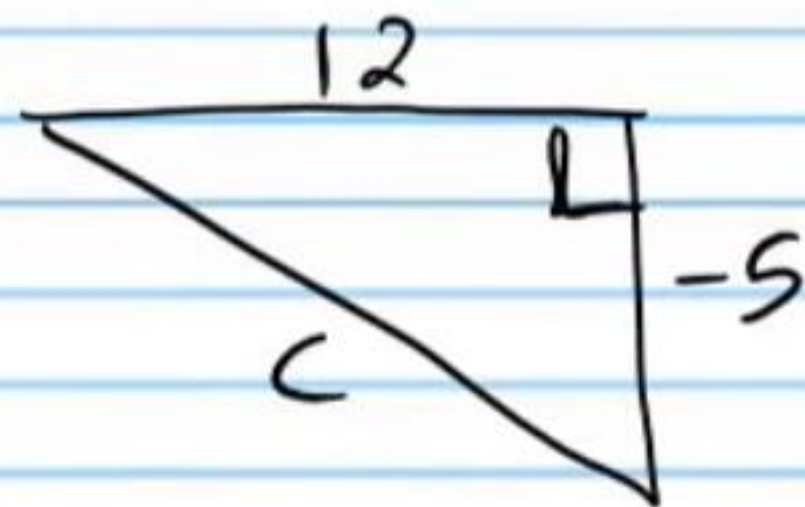
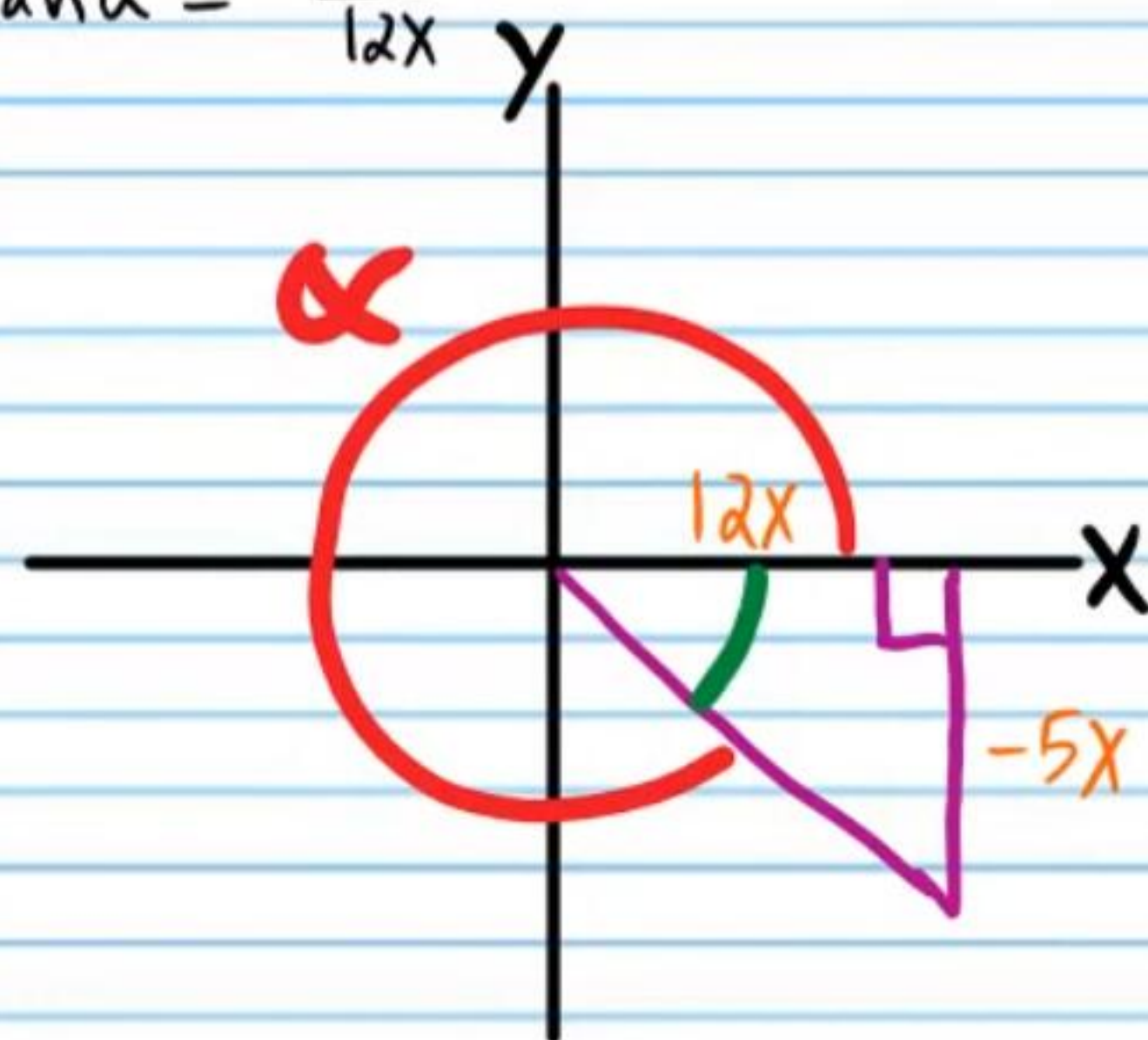
$$\text{v) } \tan \theta = \frac{1}{-2} = \boxed{-\frac{1}{2}}$$



② If $\tan \alpha = -\frac{5}{12}$, and α terminates in quadrant IV, find the values of the five remaining basic trigonometric functions of α .

$$\tan \alpha = -\frac{5x}{12x}$$

$$\text{Let } x = 1.$$



$$\begin{aligned} a^2 + b^2 &= c^2 \\ 12^2 + (-5)^2 &= c^2 \\ 144 + 25 &= c^2 \\ 169 &= c^2 \end{aligned}$$

$$\begin{aligned} \pm \sqrt{169} &= \sqrt{c^2} \\ \pm 13 &= c \end{aligned}$$

$$c = 13$$

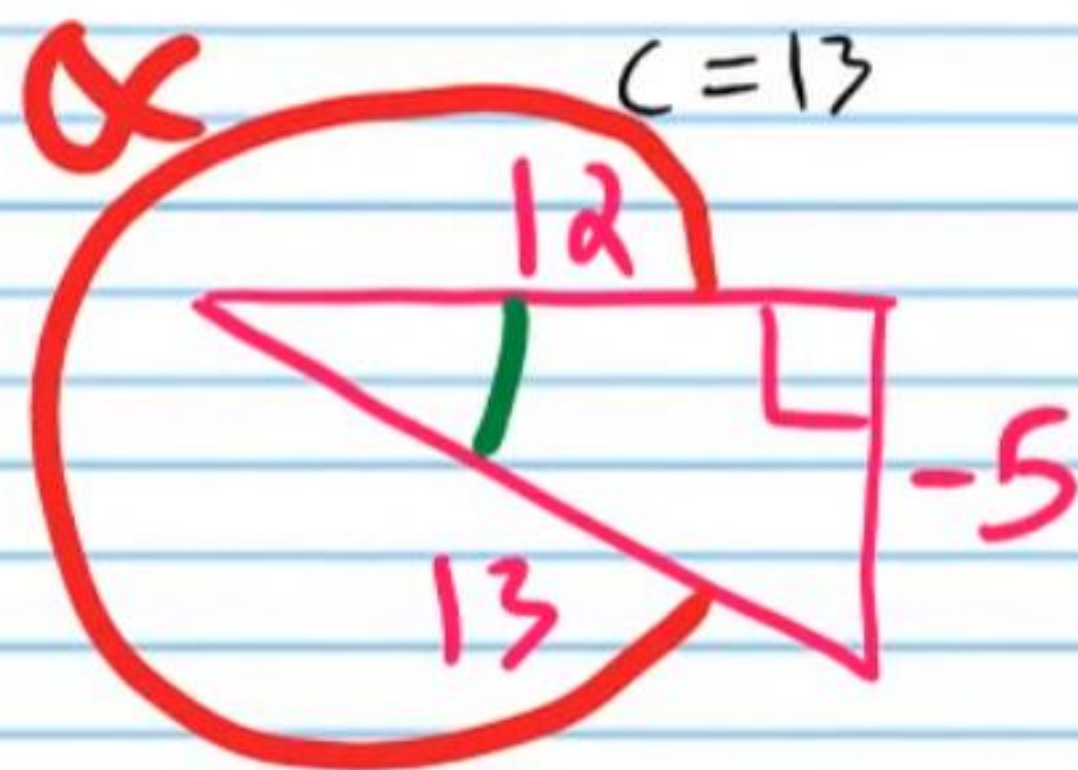
$$\text{i) } \cos \alpha = \frac{12}{13}$$

$$\text{ii) } \sin \alpha = \frac{-5}{13}$$

$$\text{iii) } \csc \alpha = \frac{13}{-5}$$

$$\text{iv) } \sec \alpha = \frac{13}{12}$$

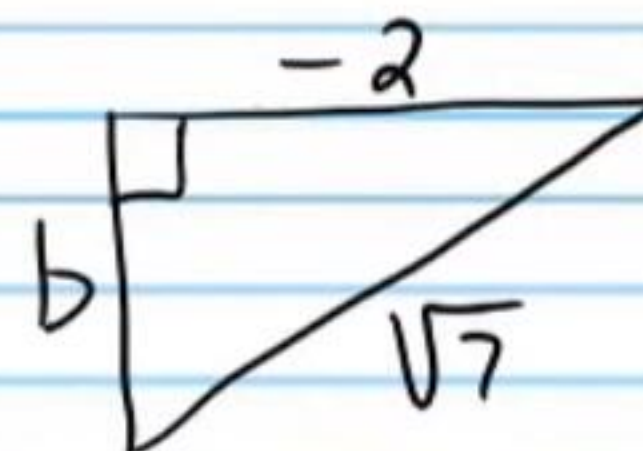
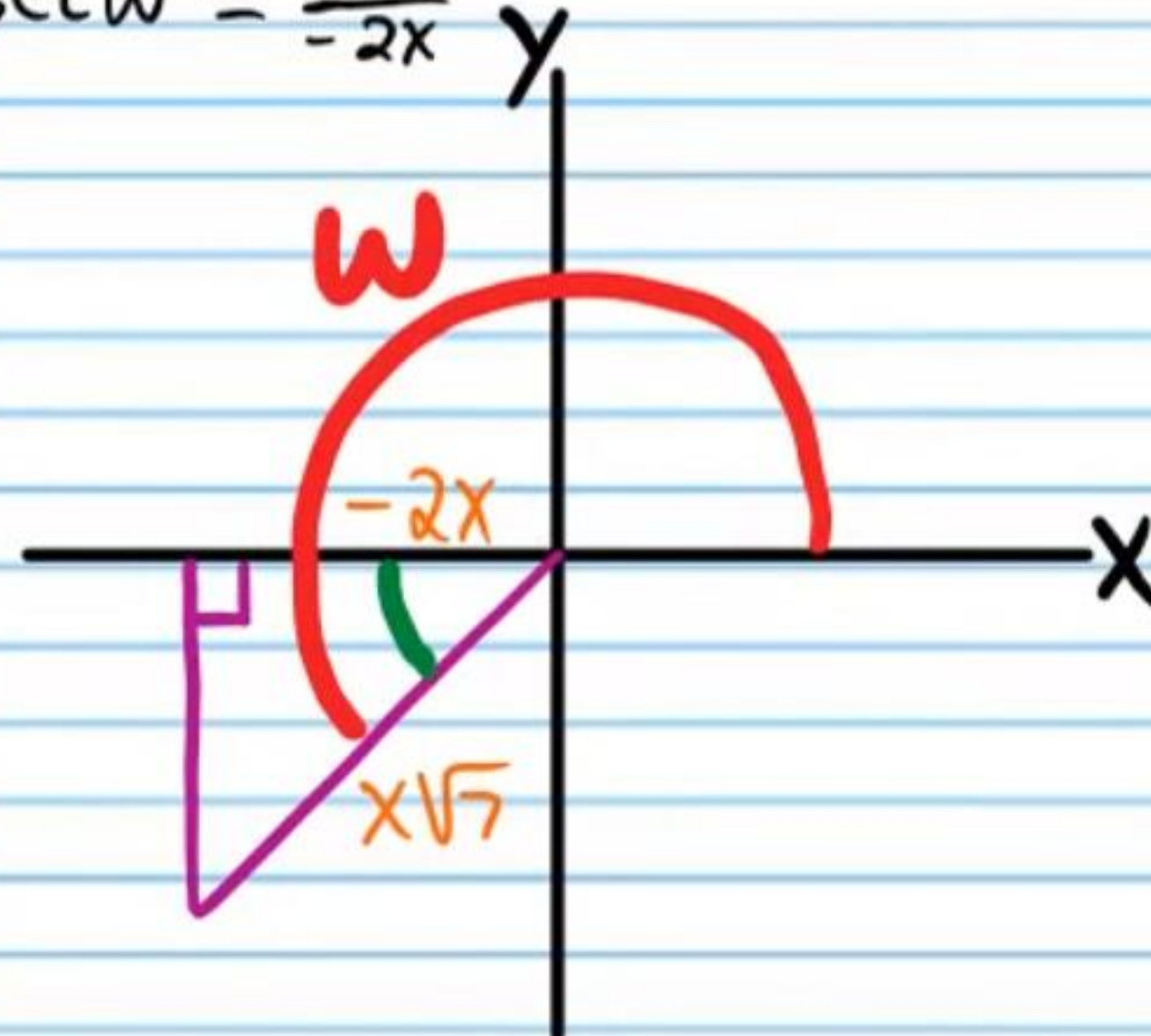
$$\text{v) } \cot \alpha = \frac{12}{-5}$$



③ If $\sec w = -\frac{\sqrt{7}}{2}$, and w terminates in quadrant III, find the values of the five remaining basic trigonometric functions of w .

$$\sec w = \frac{x\sqrt{7}}{-2x}$$

$$\text{Let } x = 1$$



$$\begin{aligned} a^2 + b^2 &= c^2 \\ (-2)^2 + b^2 &= (\sqrt{7})^2 \\ 4 + b^2 &= 7 \\ b^2 &= 3 \end{aligned}$$

$$b = \pm \sqrt{3}$$

$$b = -\sqrt{3}$$

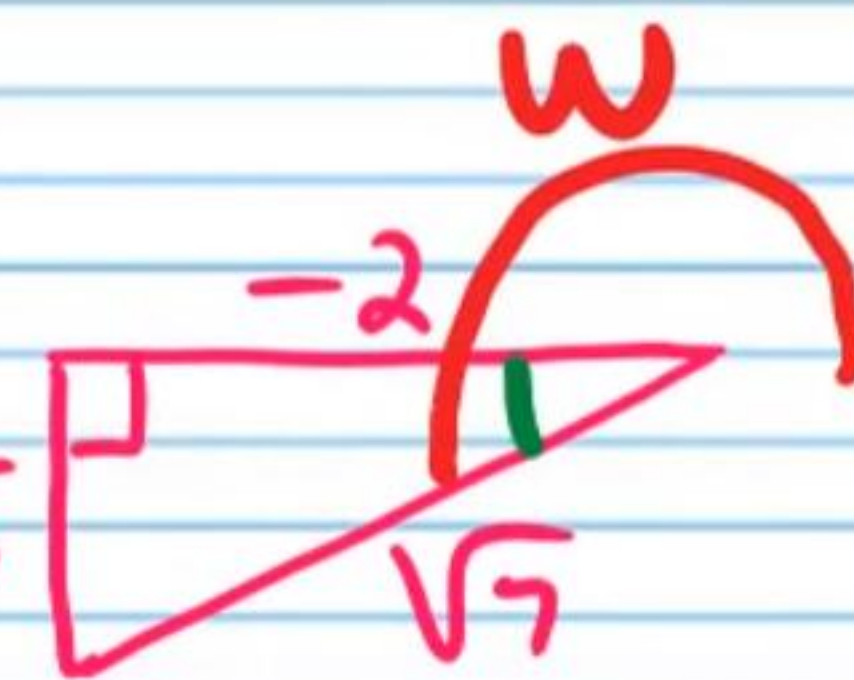
$$\text{i) } \cos w = \frac{-2}{\sqrt{7}} = \frac{-2\sqrt{7}}{7}$$

$$\text{ii) } \sin w = \frac{-\sqrt{3}}{\sqrt{7}} = \frac{-\sqrt{21}}{7}$$

$$\text{iii) } \csc w = \frac{\sqrt{7}}{-\sqrt{3}} = \frac{-\sqrt{21}}{3}$$

$$\text{iv) } \tan w = \frac{-\sqrt{3}}{-2} = \frac{\sqrt{3}}{2}$$

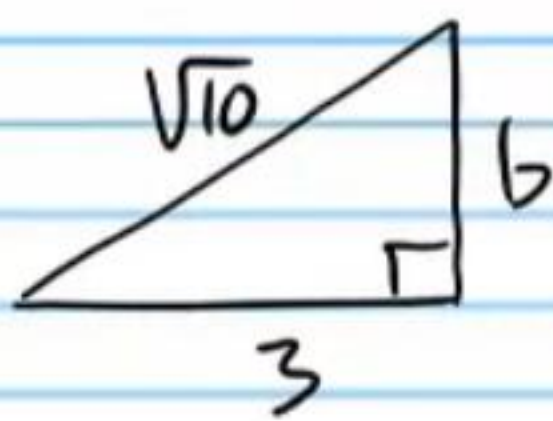
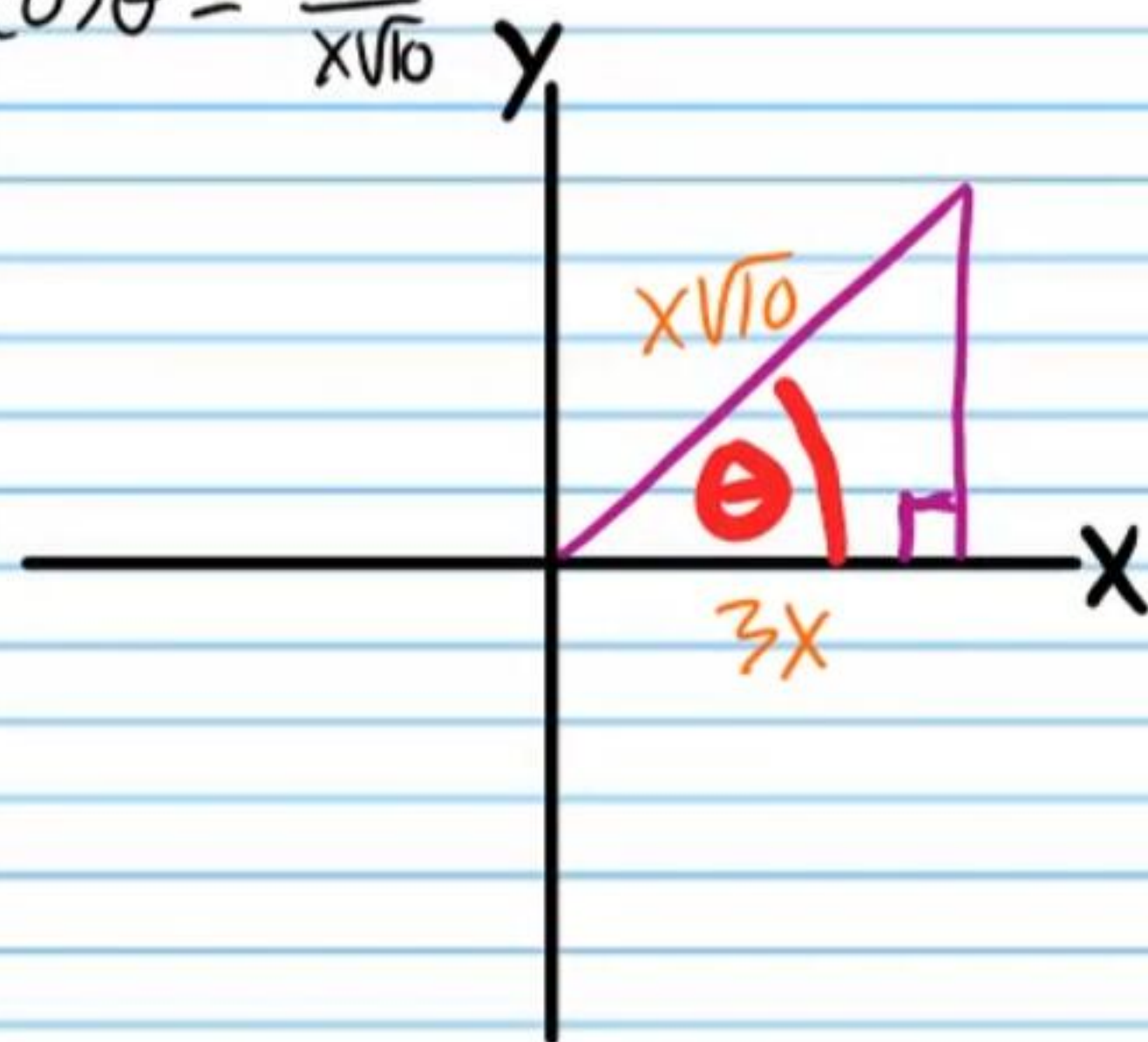
$$\text{v) } \cot w = \frac{-2}{-\sqrt{3}} = \frac{2\sqrt{3}}{3}$$



- ④ If $\cos \theta = \frac{3}{\sqrt{10}}$, and θ terminates in quadrant I, find the values of the five remaining basic trigonometric functions of θ .

Let $x=1$.

$$\cos \theta = \frac{3x}{x\sqrt{10}}$$



$$3^2 + b^2 = (\sqrt{10})^2$$

$$9 + b^2 = 10$$

$$b = \pm 1$$

$$b = 1$$

i) $\sin \theta = \frac{1}{\sqrt{10}} = \boxed{\frac{\sqrt{10}}{10}}$

ii) $\tan \theta = \boxed{\frac{1}{3}}$

iii) $\csc \theta = \frac{\sqrt{10}}{1} = \boxed{\sqrt{10}}$

iv) $\sec \theta = \boxed{\frac{\sqrt{10}}{3}}$

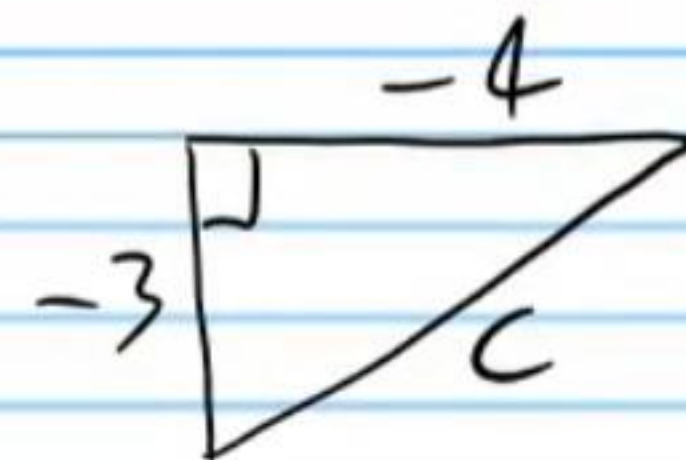
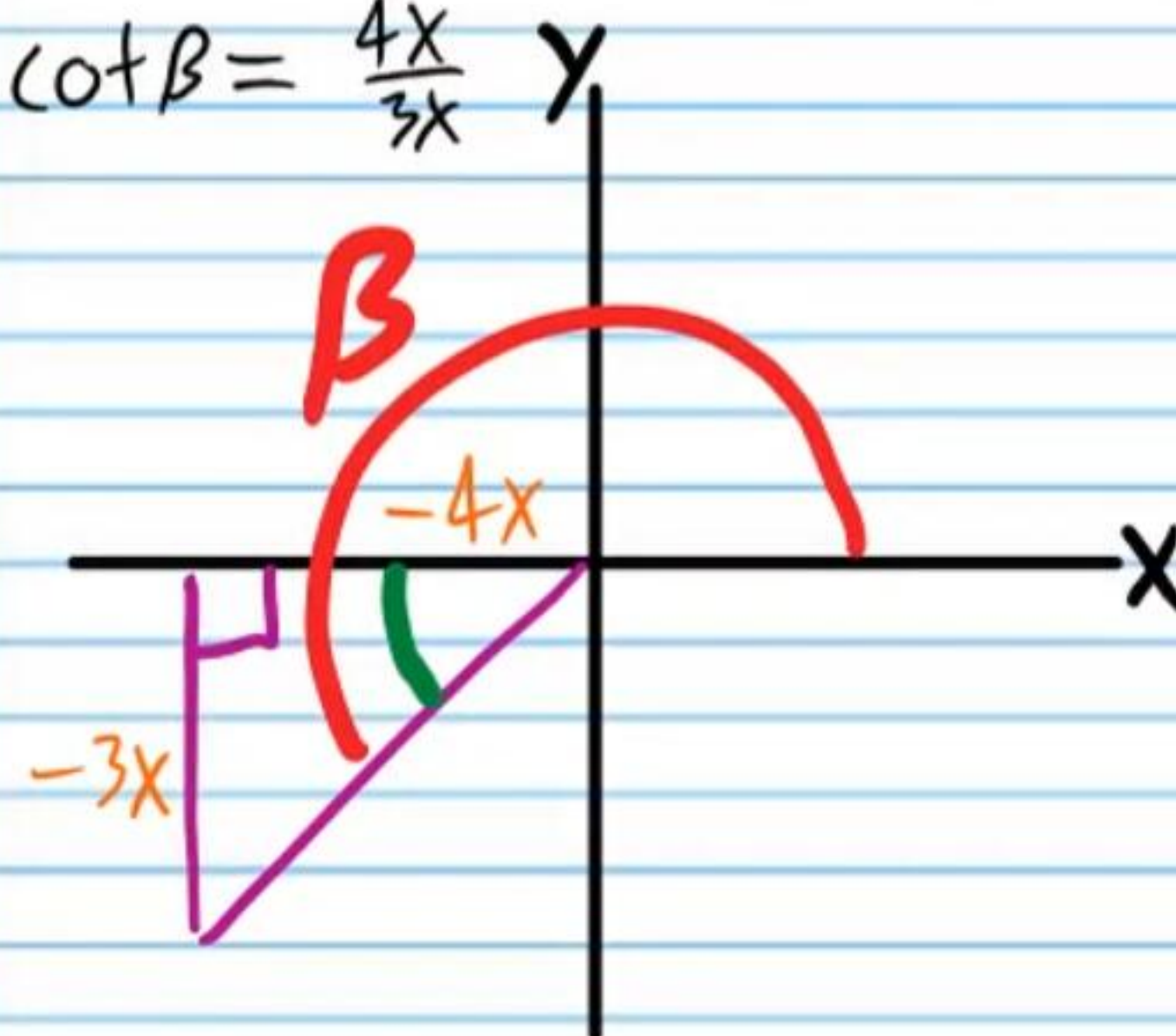
v) $\cot \theta = \frac{3}{1} = \boxed{3}$



- ⑤ If $\cot \beta = \frac{4}{3}$, and β terminates in quadrant III, find the values of the five remaining basic trigonometric functions of β .

Let $x=1$

$$\cot \beta = \frac{4x}{3x}$$



$$(-3)^2 + (-4)^2 = c^2$$

$$9 + 16 = c^2$$

$$25 = c^2$$

$$c = \pm 5$$

$$c = 5$$

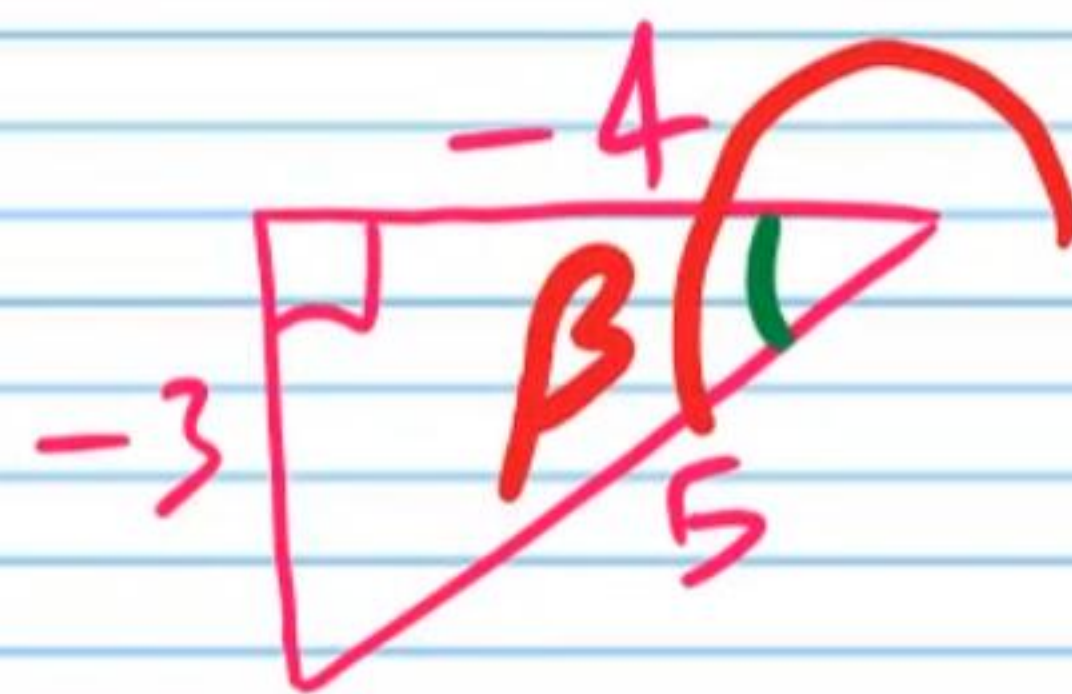
i) $\cos \beta = \boxed{-\frac{4}{5}}$

ii) $\sin \beta = \boxed{-\frac{3}{5}}$

iii) $\csc \beta = \boxed{\frac{5}{-3}}$

iv) $\sec \beta = \boxed{\frac{5}{-4}}$

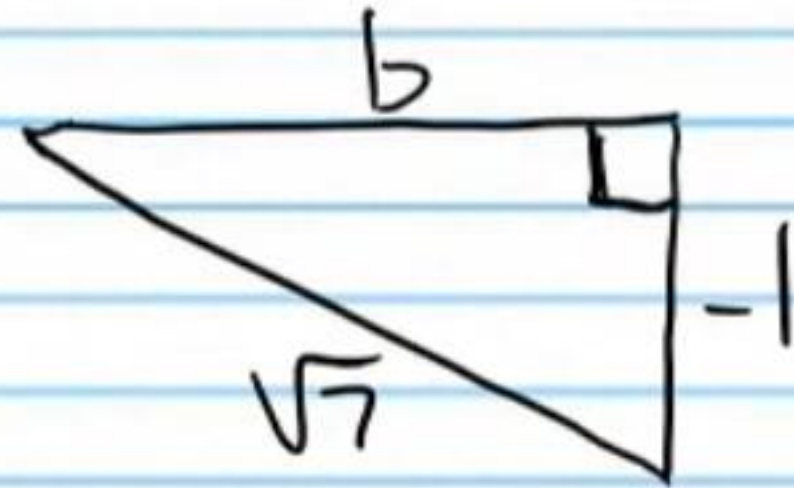
v) $\tan \beta = \frac{-3}{-4} = \boxed{\frac{3}{4}}$



⑥ If $\csc W = -\sqrt{7}$, and W terminates in quadrant IV, find the values of the five remaining basic trigonometric functions of W .

$$\csc W = \frac{x\sqrt{7}}{-x}$$

$$\text{Let } x = 1.$$

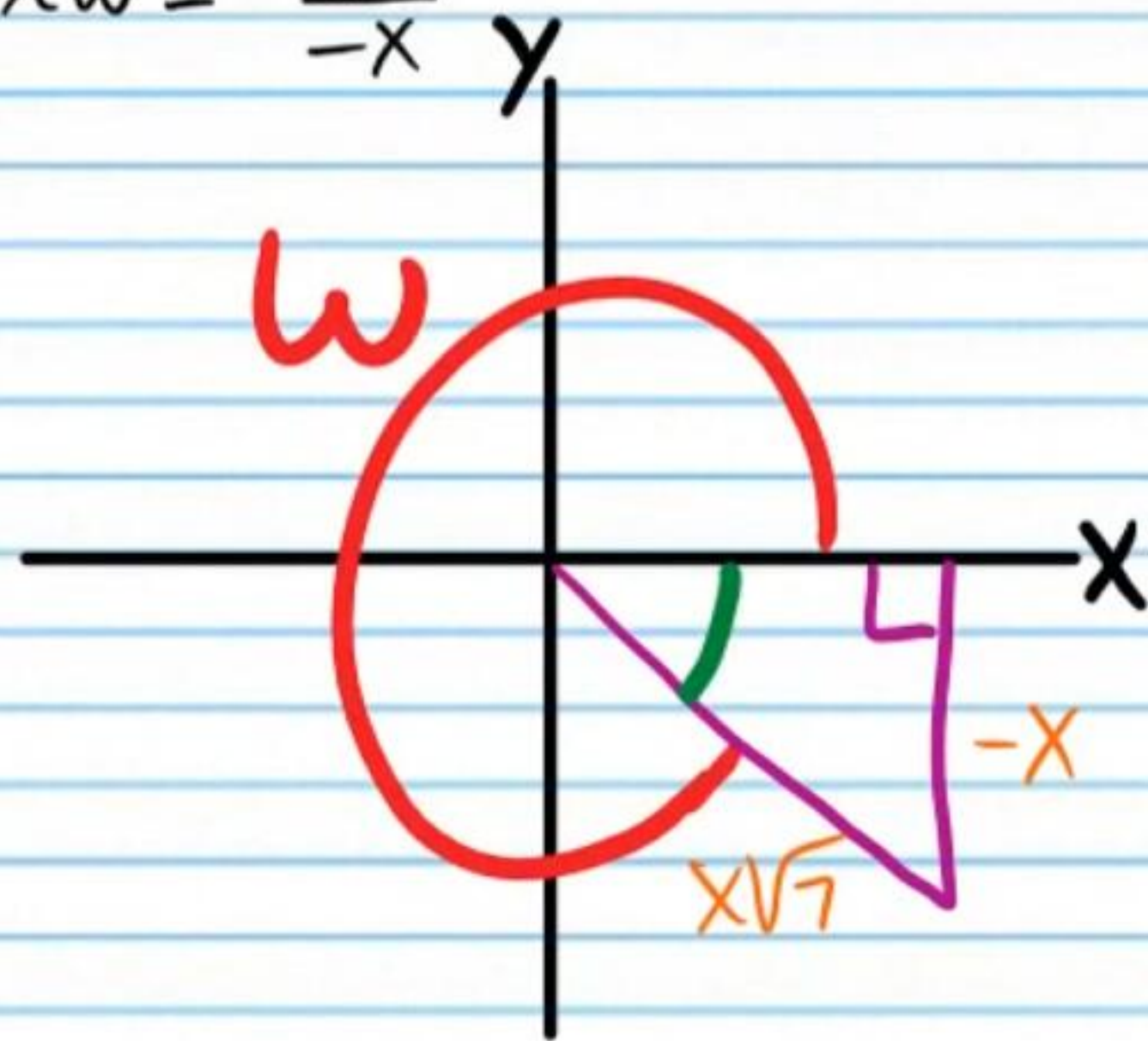


$$(-1)^2 + b^2 = (\sqrt{7})^2$$

$$1 + b^2 = 7$$

$$b = \pm\sqrt{6}$$

$$b = \sqrt{6}$$



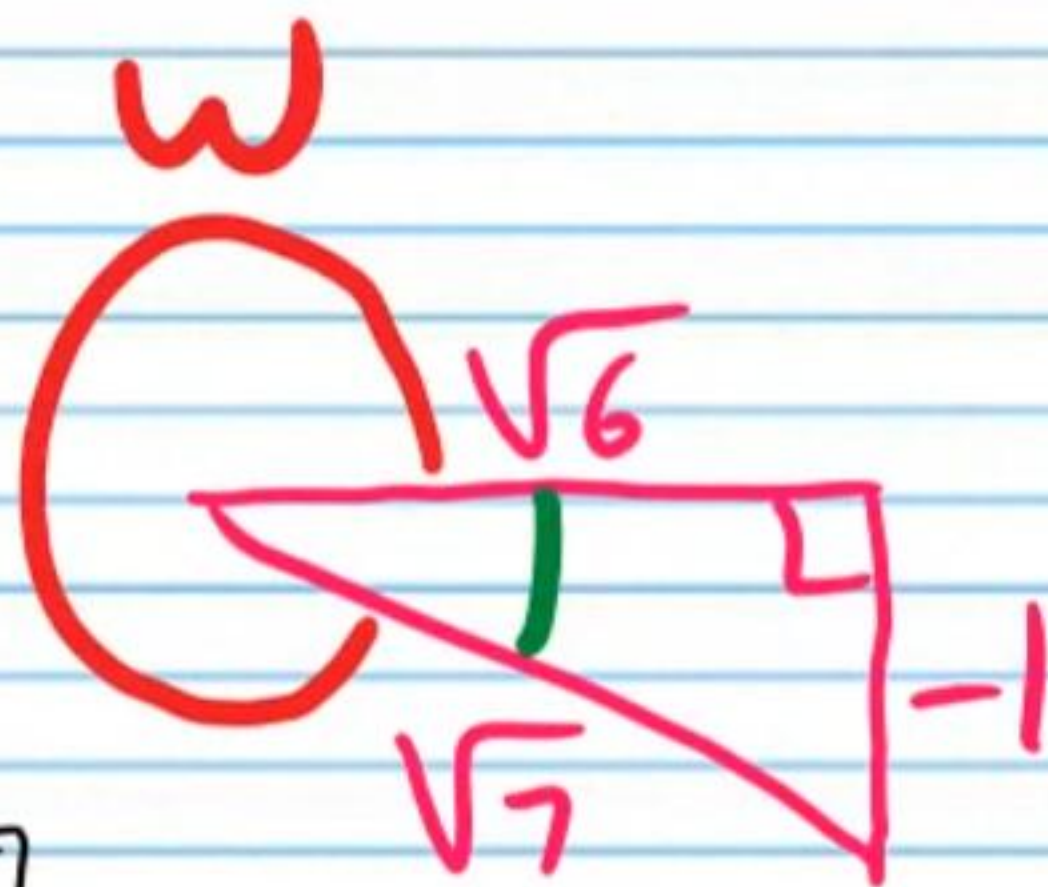
$$\text{i) } \cos W = \frac{\sqrt{6}}{\sqrt{7}} = \boxed{\frac{\sqrt{42}}{7}}$$

$$\text{ii) } \sin W = -\frac{1}{\sqrt{7}} = \boxed{-\frac{\sqrt{7}}{7}}$$

$$\text{iii) } \sec W = \frac{\sqrt{7}}{\sqrt{6}} = \boxed{\frac{\sqrt{42}}{6}}$$

$$\text{iv) } \tan W = \frac{-1}{\sqrt{6}} = \boxed{-\frac{\sqrt{6}}{6}}$$

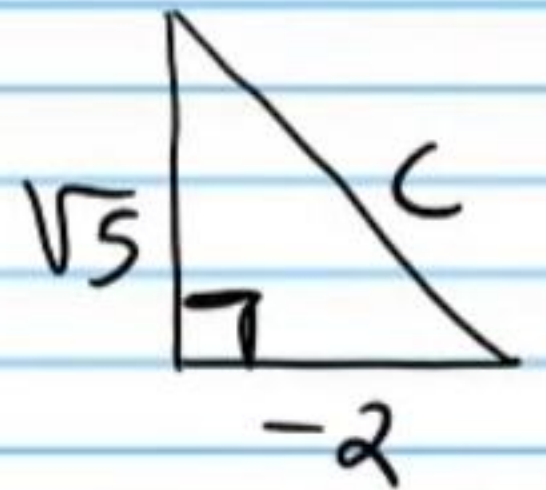
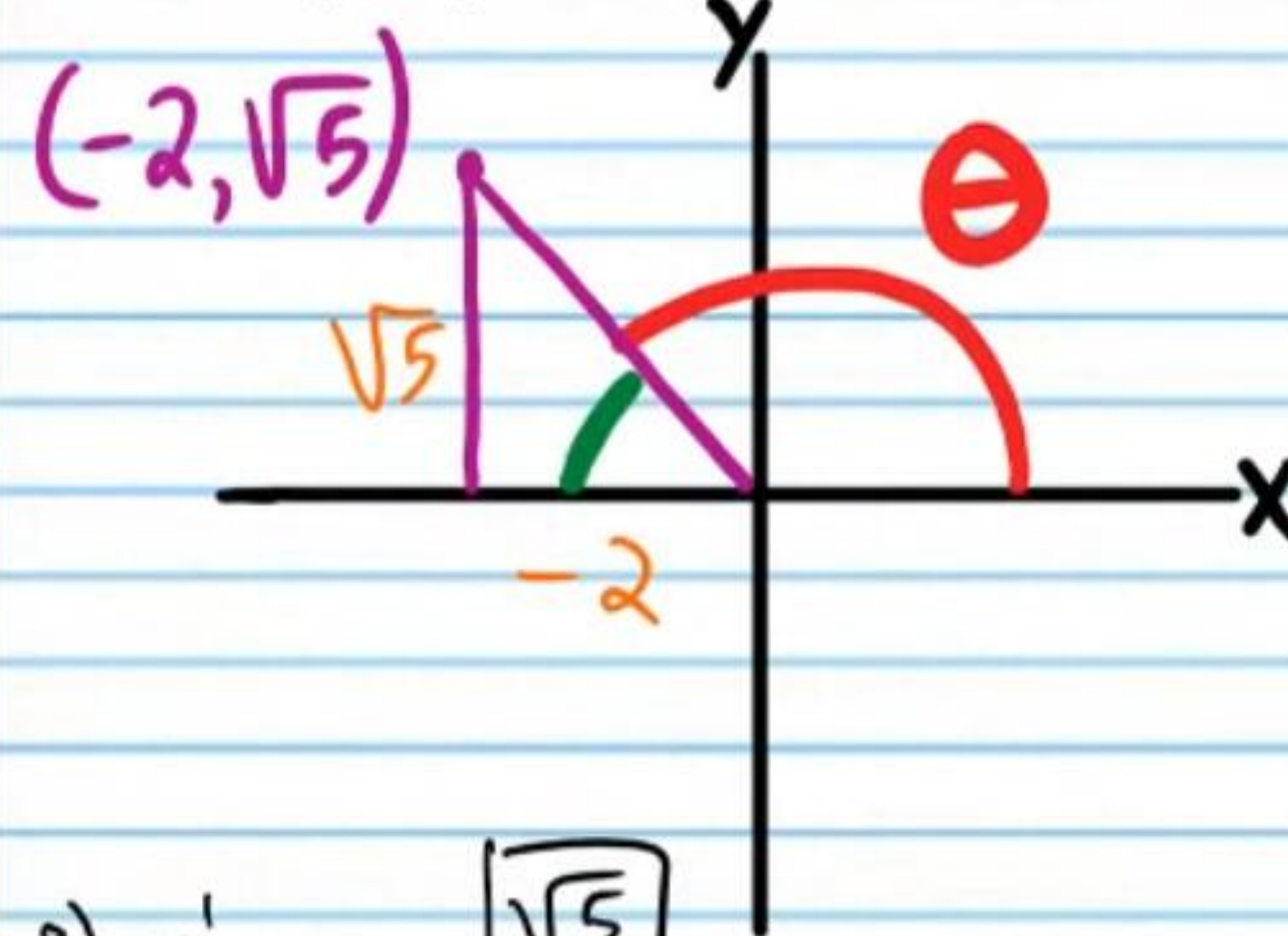
$$\text{v) } \cot W = \frac{\sqrt{6}}{-1} = \boxed{-\sqrt{6}}$$



Finding Trigonometric Values Given A Point on the Terminal Side of the Angle

The coordinates of points are given below. If each of these points lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

⑦ $(-2, \sqrt{5})$



$$(\sqrt{5})^2 + (-2)^2 = c^2$$

$$5 + 4 = c^2$$

$$c = \pm 3$$

$$c = 3$$

$$\text{i) } \sin \theta = \boxed{\frac{\sqrt{5}}{3}}$$

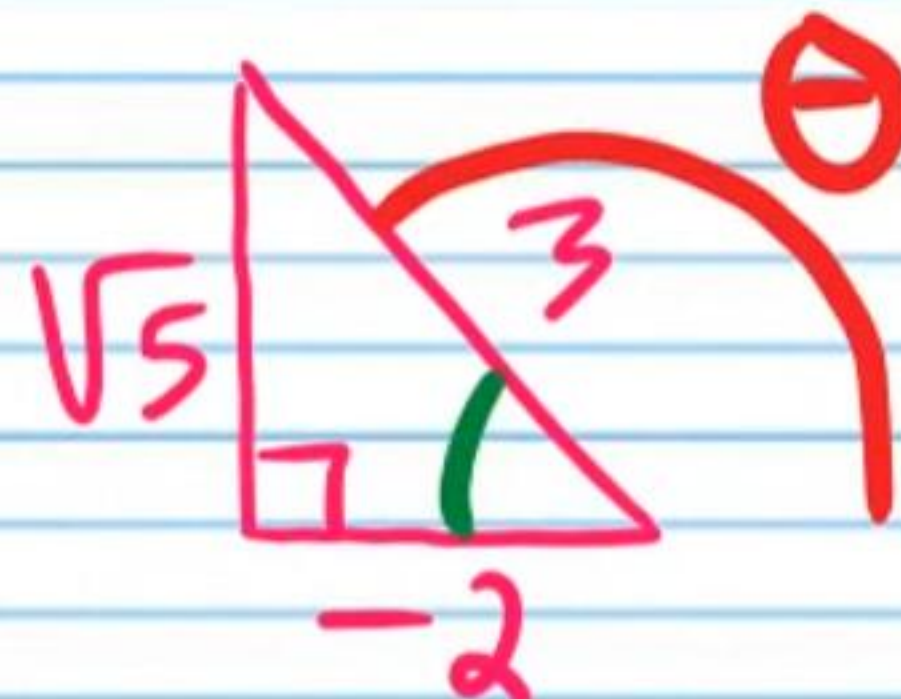
$$\text{ii) } \cos \theta = \boxed{\frac{-2}{3}}$$

$$\text{iii) } \tan \theta = \boxed{\frac{\sqrt{5}}{-2}}$$

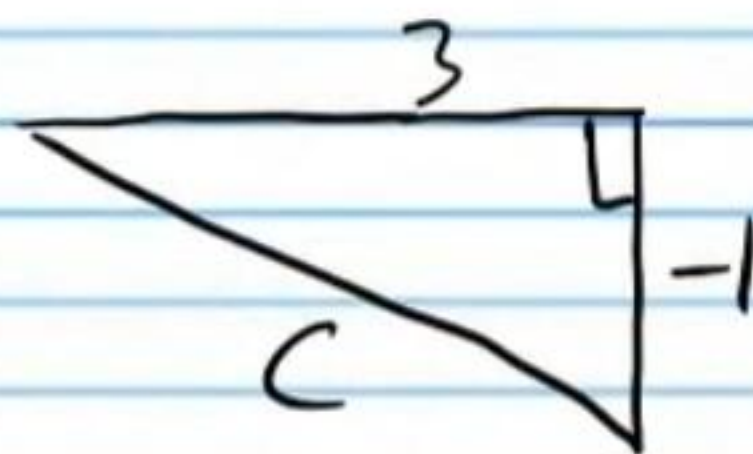
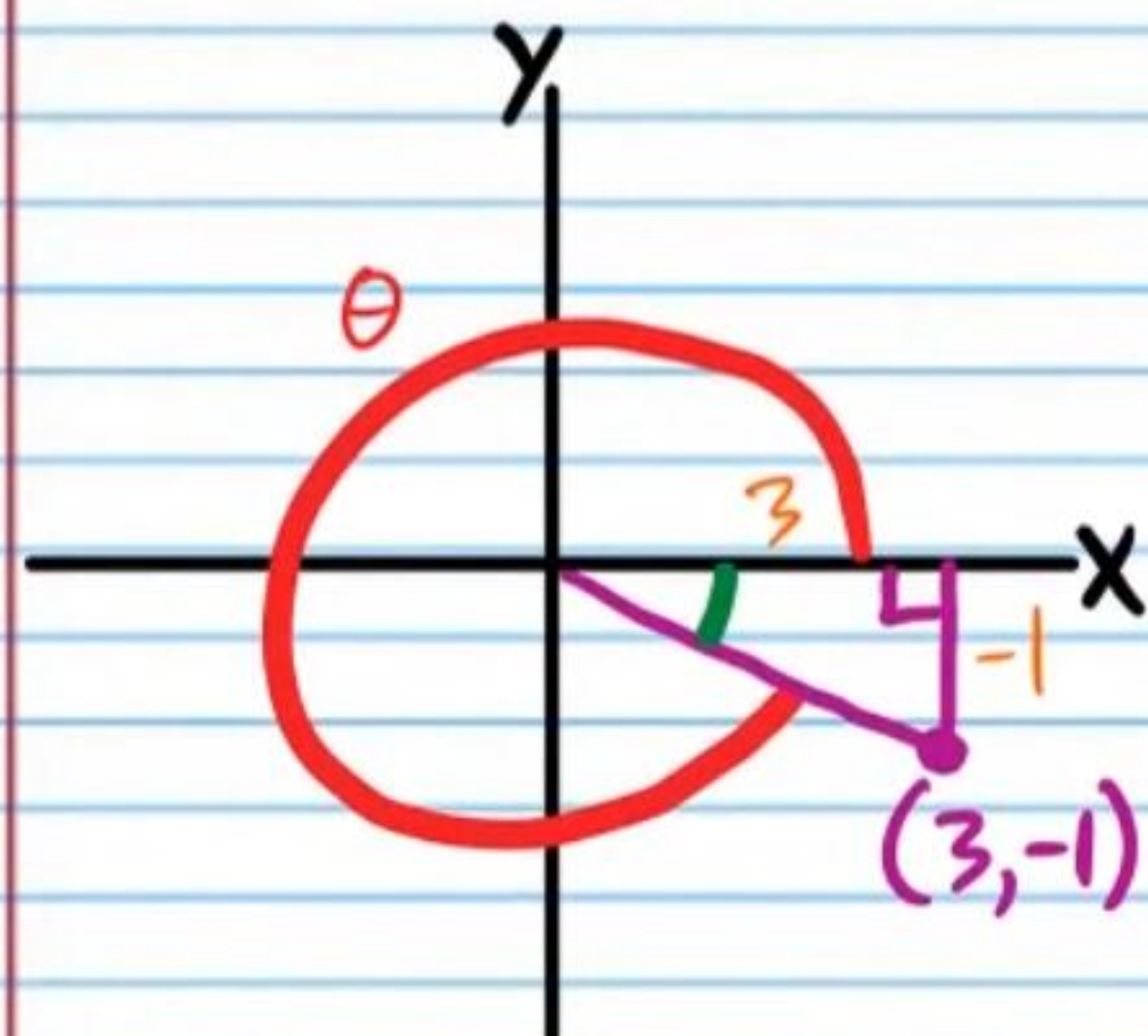
$$\text{iv) } \csc \theta = \frac{3}{\sqrt{5}} = \boxed{\frac{3\sqrt{5}}{5}}$$

$$\text{v) } \sec \theta = \boxed{\frac{3}{-2}}$$

$$\text{vi) } \cot \theta = \frac{-2}{\sqrt{5}} = \boxed{-\frac{2\sqrt{5}}{5}}$$



⑧ $(3, -1)$



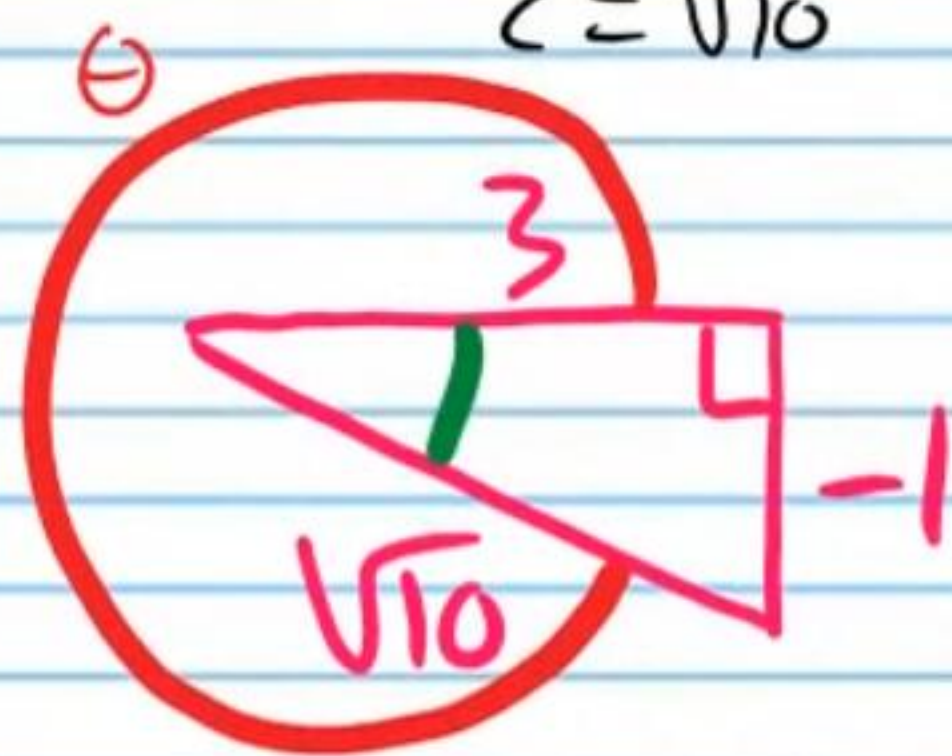
$$3^2 + (-1)^2 = C^2$$

$$9 + 1 = C^2$$

$$10 = C^2$$

$$C = \pm\sqrt{10}$$

$$C = \sqrt{10}$$



i) $\sin \theta = \frac{-1}{\sqrt{10}} = \boxed{-\frac{\sqrt{10}}{10}}$

ii) $\cos \theta = \frac{3}{\sqrt{10}} = \boxed{\frac{3\sqrt{10}}{10}}$

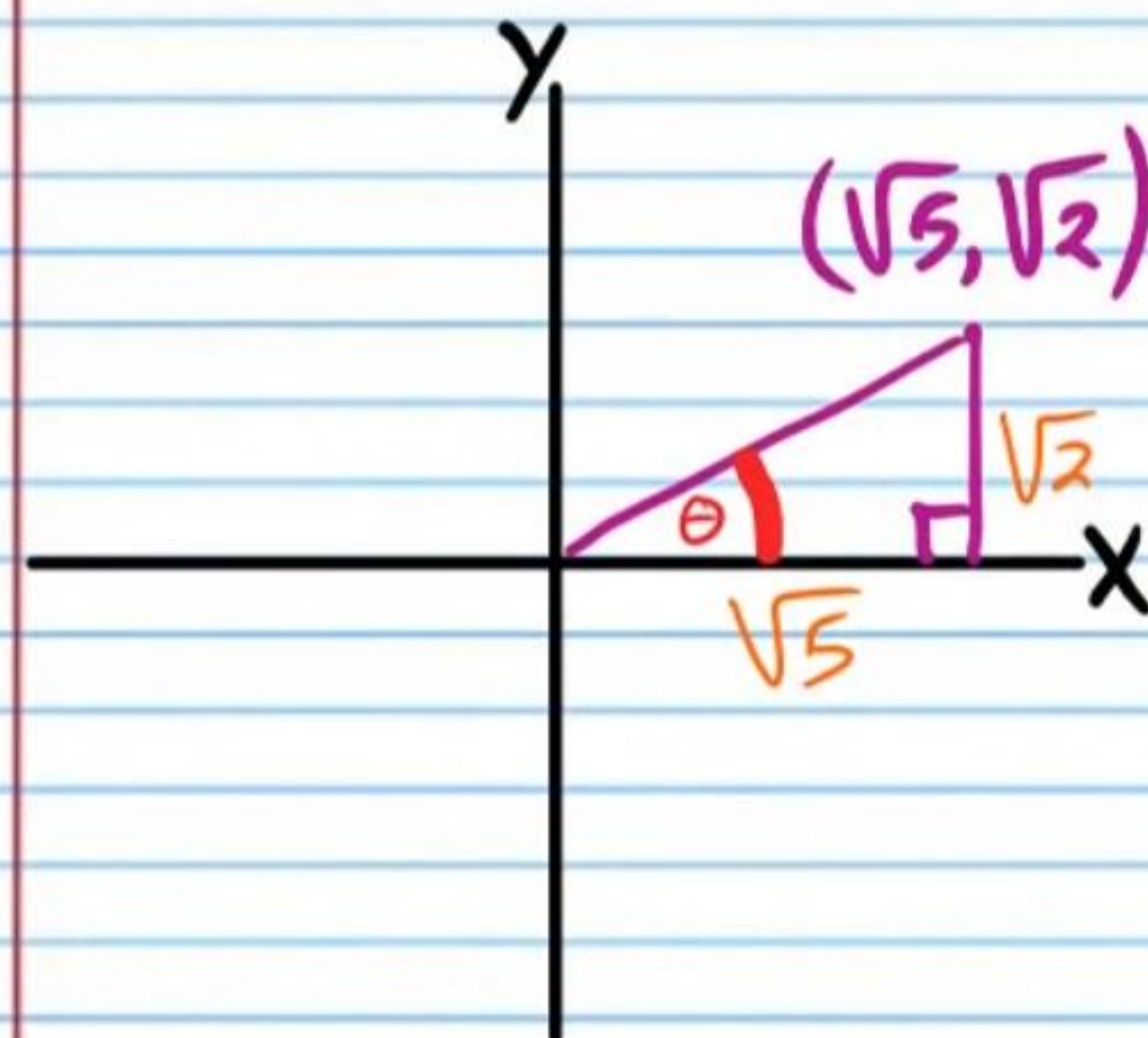
iii) $\tan \theta = \boxed{-\frac{1}{3}}$

iv) $\csc \theta = \frac{\sqrt{10}}{-1} = \boxed{-\sqrt{10}}$

v) $\sec \theta = \boxed{\frac{\sqrt{10}}{3}}$

vi) $\cot \theta = \frac{3}{-1} = \boxed{-3}$

⑨ $(\sqrt{5}, \sqrt{2})$



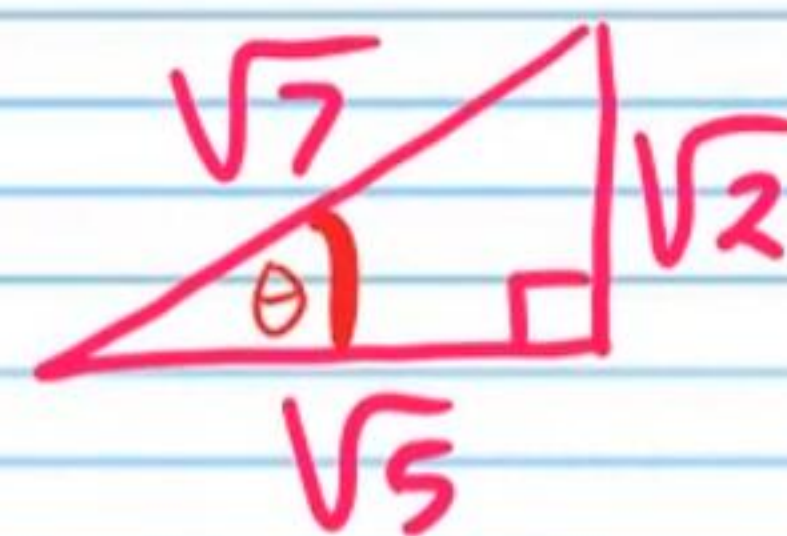
$$(\sqrt{5})^2 + (\sqrt{2})^2 = C^2$$

$$5 + 2 = C^2$$

$$7 = C^2$$

$$\pm\sqrt{7} = C$$

$$C = \sqrt{7}$$



i) $\sin \theta = \frac{\sqrt{2}}{\sqrt{7}} = \boxed{\frac{\sqrt{14}}{7}}$

ii) $\cos \theta = \frac{\sqrt{5}}{\sqrt{7}} = \boxed{\frac{\sqrt{35}}{7}}$

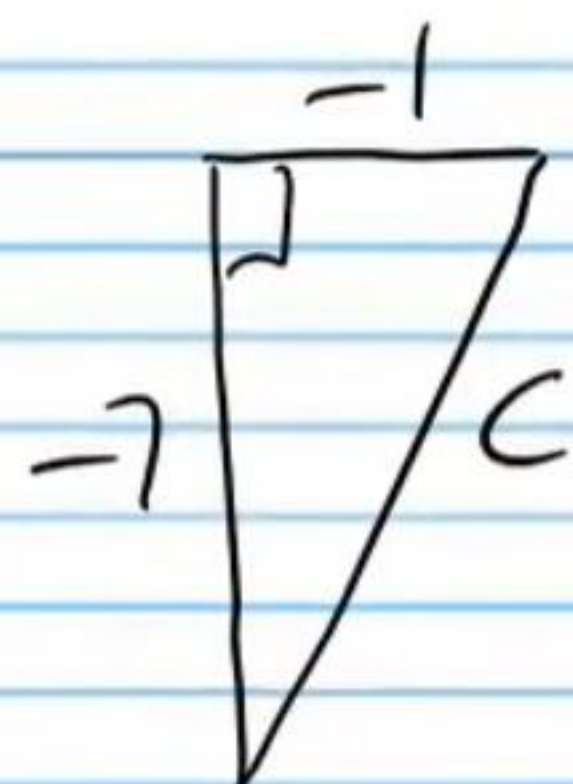
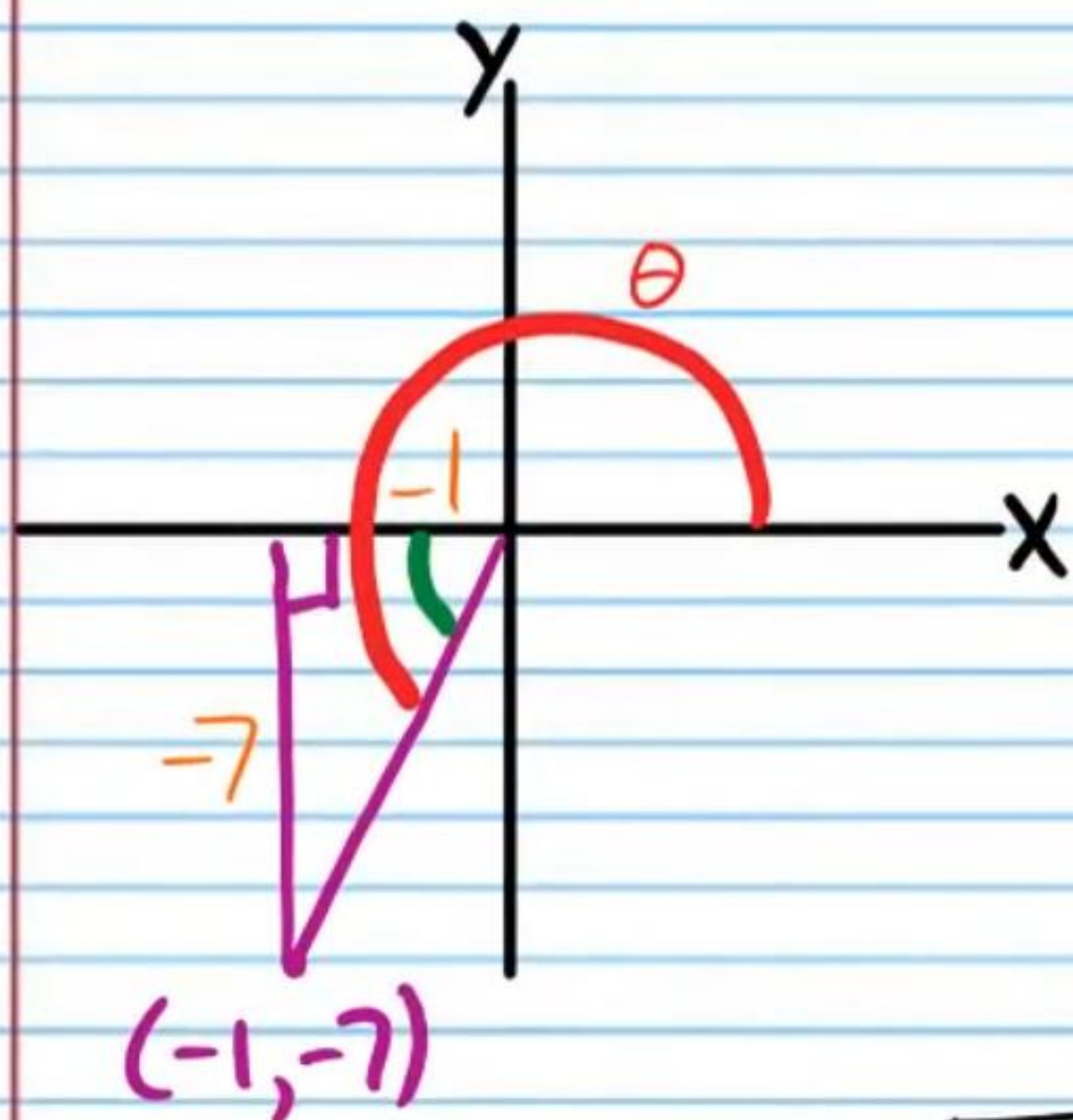
iii) $\tan \theta = \frac{\sqrt{2}}{\sqrt{5}} = \boxed{\frac{\sqrt{10}}{5}}$

iv) $\csc \theta = \frac{\sqrt{7}}{\sqrt{2}} = \boxed{\frac{\sqrt{14}}{2}}$

v) $\sec \theta = \frac{\sqrt{7}}{\sqrt{5}} = \boxed{\frac{\sqrt{35}}{5}}$

vi) $\cot \theta = \frac{\sqrt{5}}{\sqrt{2}} = \boxed{\frac{\sqrt{10}}{2}}$

⑩ $(-1, -7)$



$$\begin{aligned} (-7)^2 + (-1)^2 &= c^2 \\ 49 + 1 &= c^2 \\ 50 &= c^2 \\ \pm\sqrt{50} &= c \end{aligned}$$

i) $\sin \theta = \frac{-7}{5\sqrt{2}} = \frac{-7\sqrt{2}}{5 \cdot 2} = \boxed{-\frac{7\sqrt{2}}{10}}$

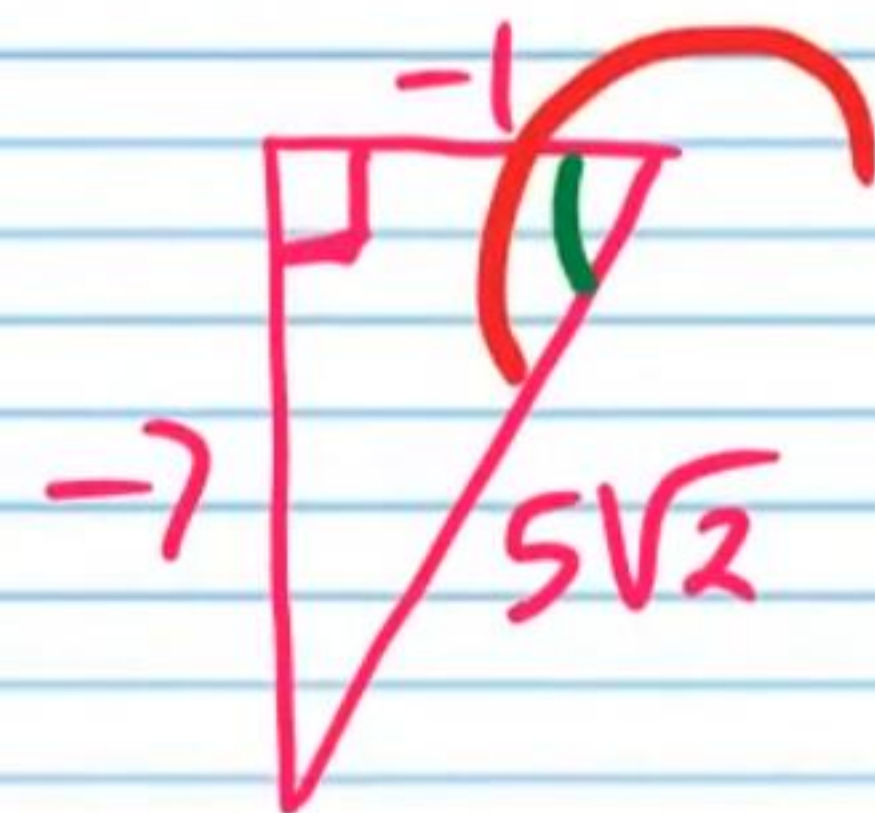
ii) $\cos \theta = \frac{-1}{5\sqrt{2}} = \frac{-\sqrt{2}}{5 \cdot 2} = \boxed{-\frac{\sqrt{2}}{10}}$

iii) $\tan \theta = \frac{-7}{-1} = \boxed{7}$

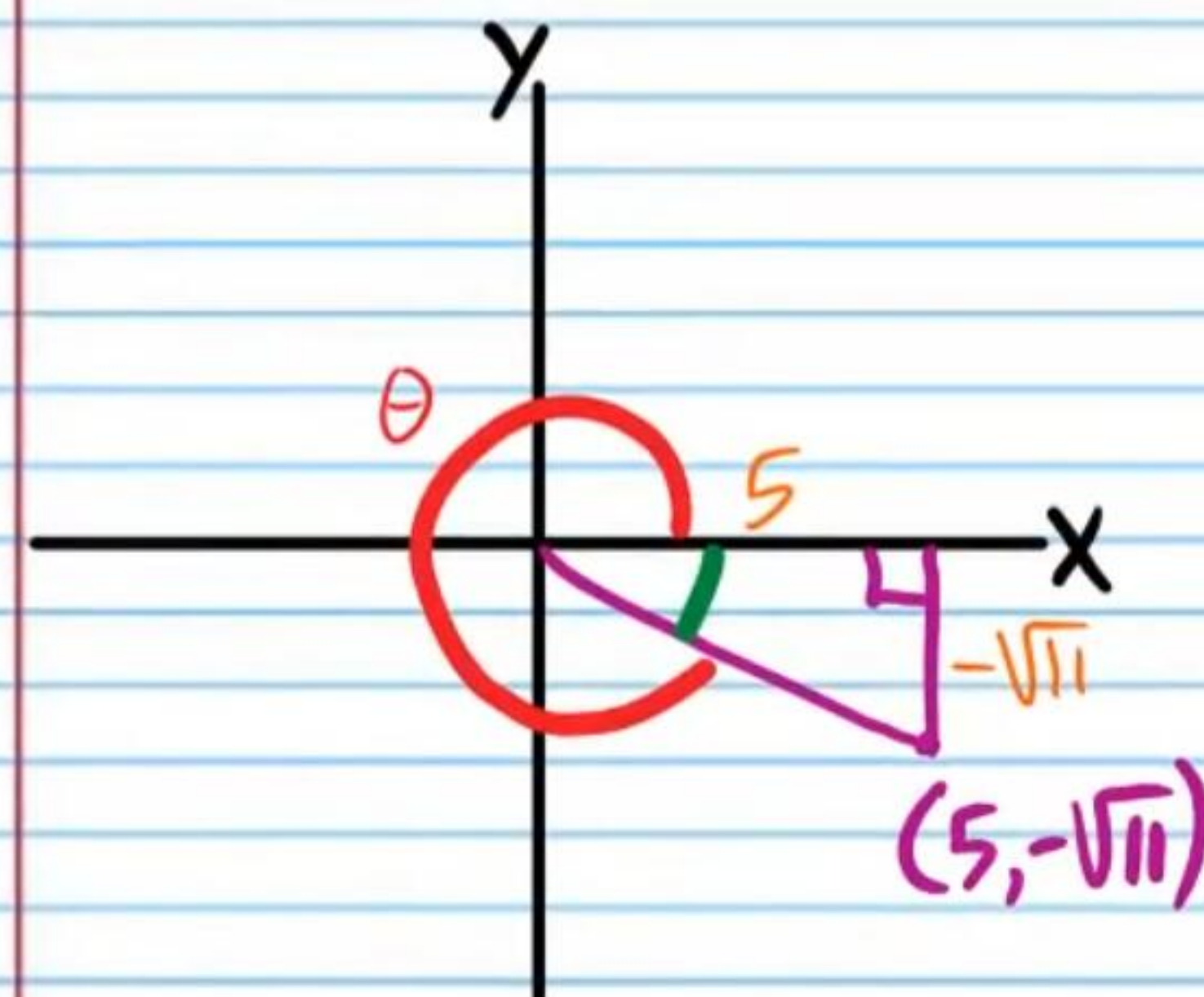
iv) $\csc \theta = \frac{5\sqrt{2}}{-7} = \boxed{-\frac{5\sqrt{2}}{7}}$

v) $\sec \theta = \frac{5\sqrt{2}}{-1} = \boxed{-5\sqrt{2}}$

vi) $\cot \theta = \frac{-1}{-7} = \boxed{\frac{1}{7}}$



• ⑪ $(5, -\sqrt{11})$



$$\begin{aligned} 5^2 + (-\sqrt{11})^2 &= c^2 \\ 25 + 11 &= c^2 \\ 36 &= c^2 \\ \pm 6 &= c \\ c &= 6 \end{aligned}$$

i) $\sin \theta = \frac{-\sqrt{11}}{6} = \boxed{-\frac{\sqrt{11}}{6}}$

ii) $\cos \theta = \frac{5}{6} = \boxed{\frac{5}{6}}$

iii) $\tan \theta = \frac{-\sqrt{11}}{5} = \boxed{-\frac{\sqrt{11}}{5}}$

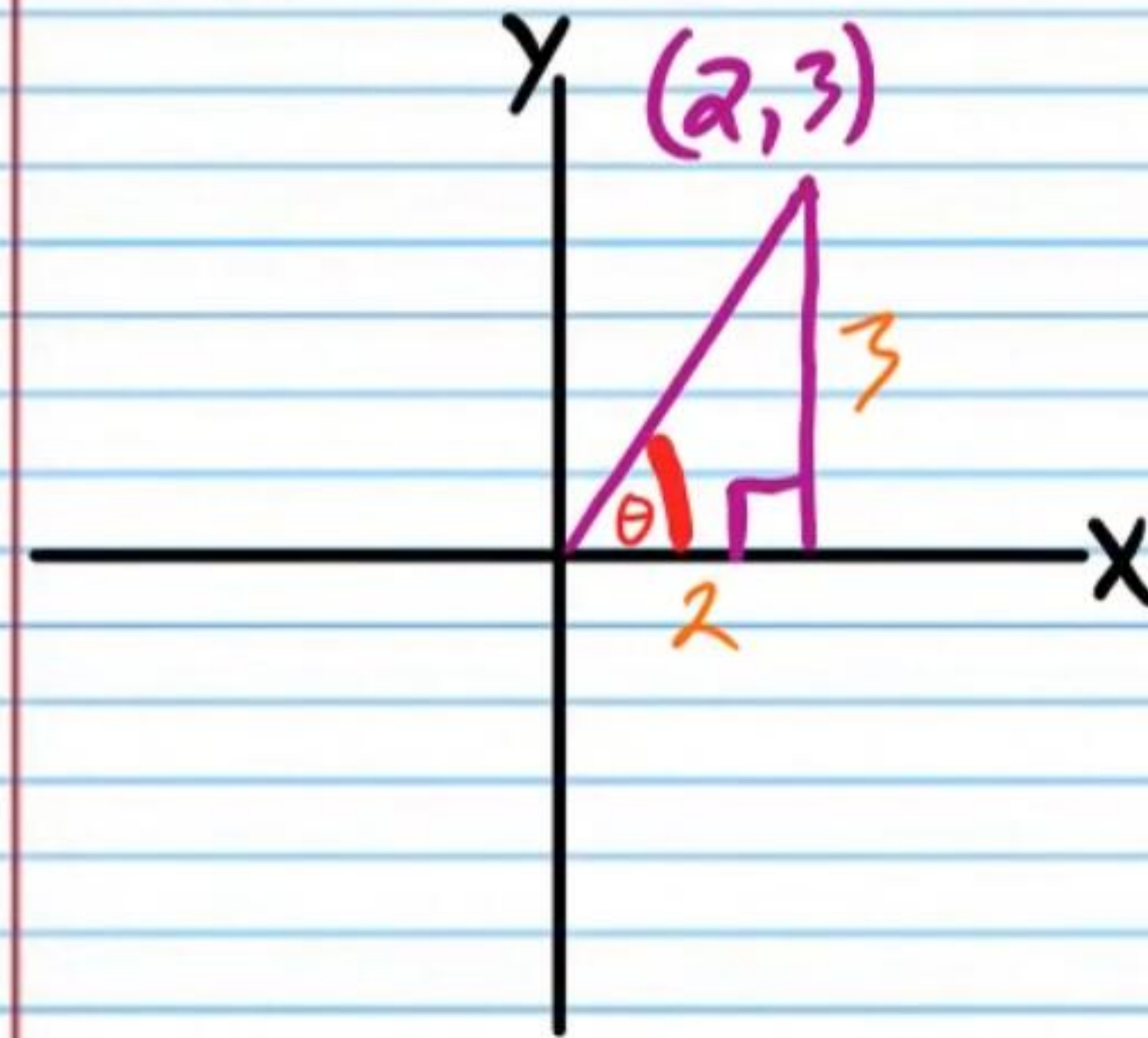
iv) $\csc \theta = \frac{6}{-\sqrt{11}} = \boxed{-\frac{6\sqrt{11}}{11}}$

v) $\sec \theta = \frac{6}{5} = \boxed{\frac{6}{5}}$

vi) $\cot \theta = \frac{5}{-\sqrt{11}} = \boxed{-\frac{5\sqrt{11}}{11}}$

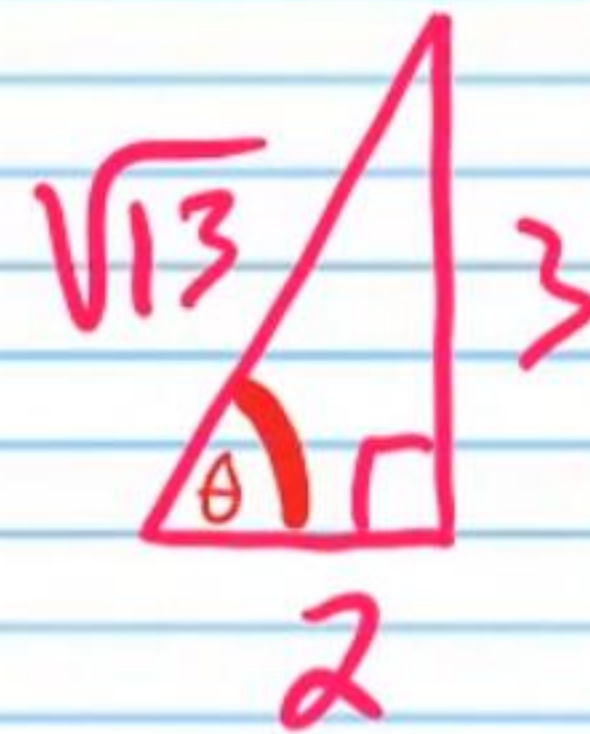


⑫ (2, 3)

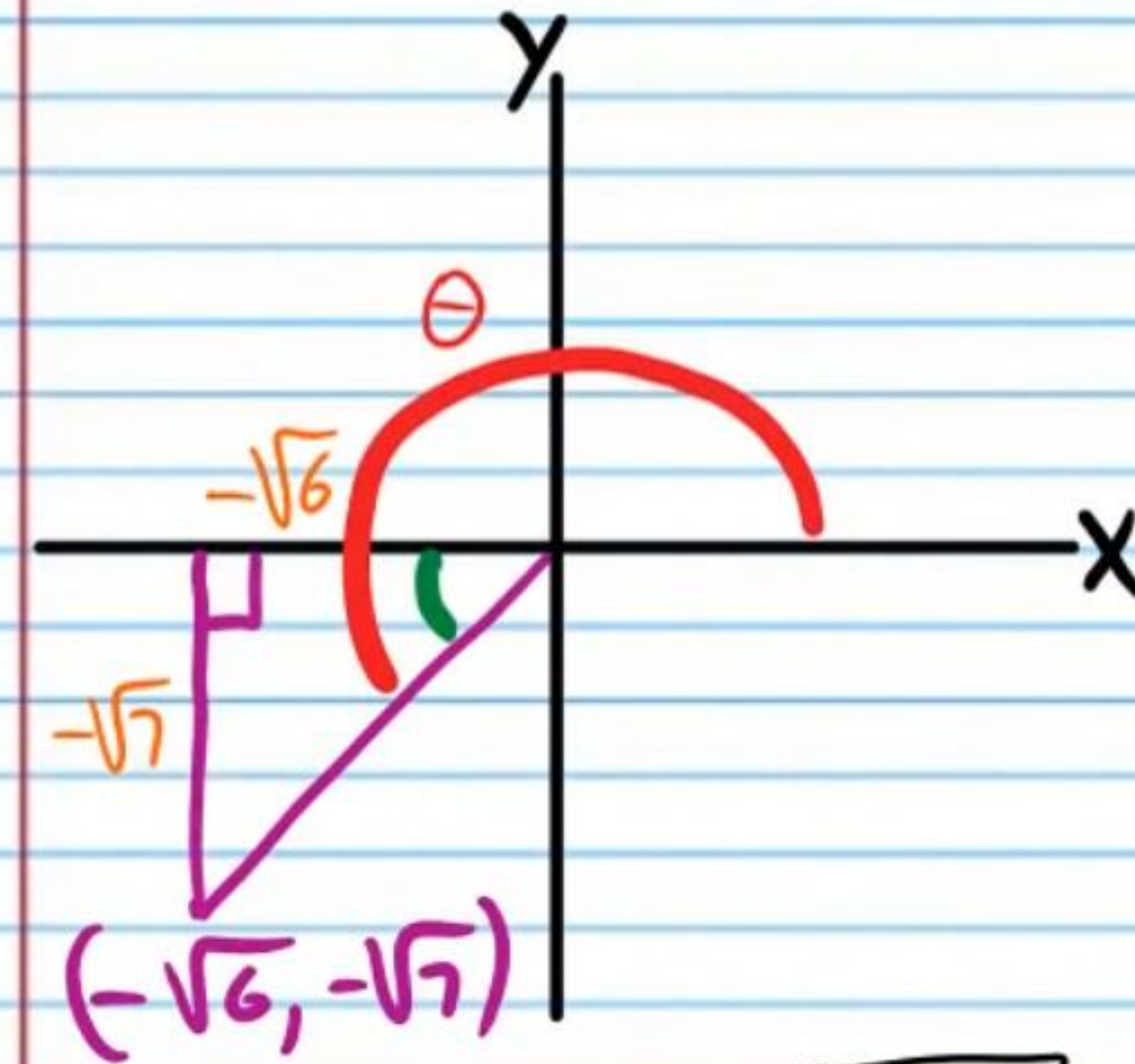


$$\begin{aligned} 3^2 + 2^2 &= c^2 \\ 9 + 4 &= c^2 \\ c &= \sqrt{13} \end{aligned}$$

$$\begin{aligned} \text{i) } \sin \theta &= \frac{3}{\sqrt{13}} = \boxed{\frac{3\sqrt{13}}{13}} \\ \text{ii) } \cos \theta &= \frac{2}{\sqrt{13}} = \boxed{\frac{2\sqrt{13}}{13}} \\ \text{iii) } \tan \theta &= \boxed{\frac{3}{2}} \\ \text{iv) } \csc \theta &= \boxed{\frac{\sqrt{13}}{3}} \\ \text{v) } \sec \theta &= \boxed{\frac{\sqrt{13}}{2}} \\ \text{vi) } \cot \theta &= \boxed{\frac{2}{3}} \end{aligned}$$

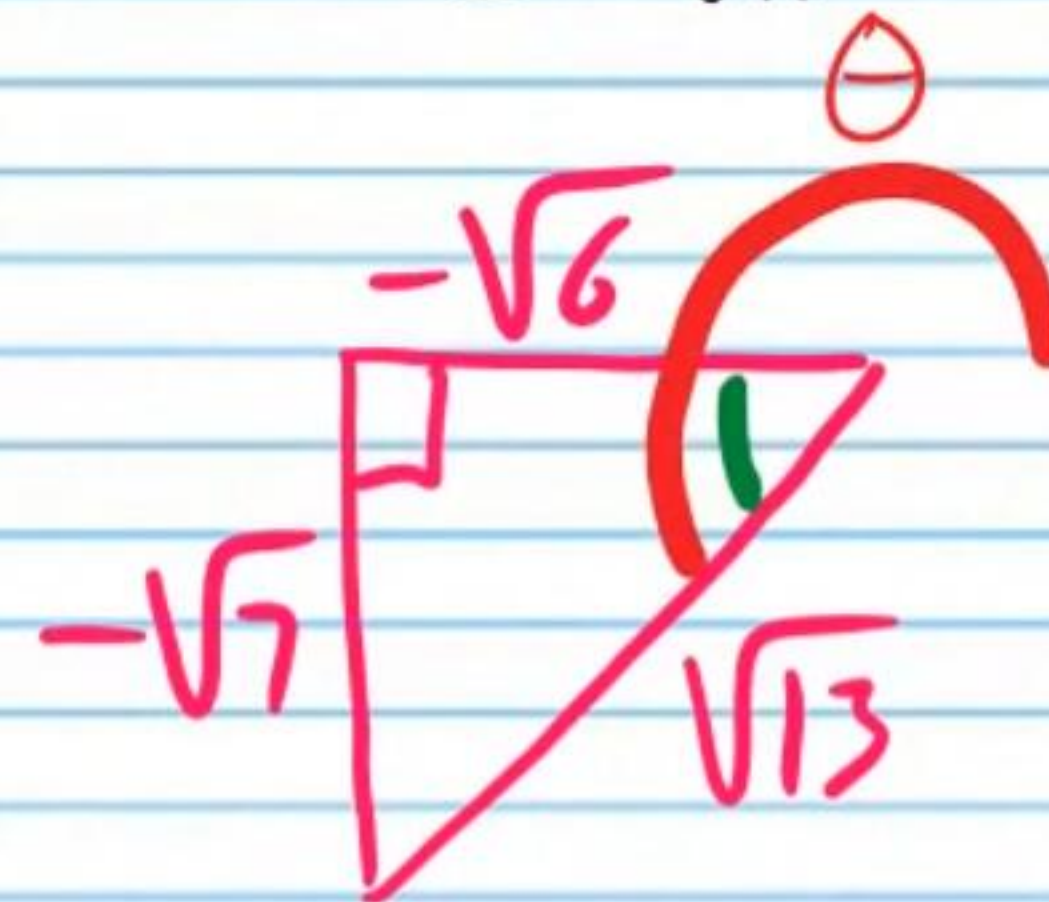


⑬ $(-\sqrt{6}, -\sqrt{7})$

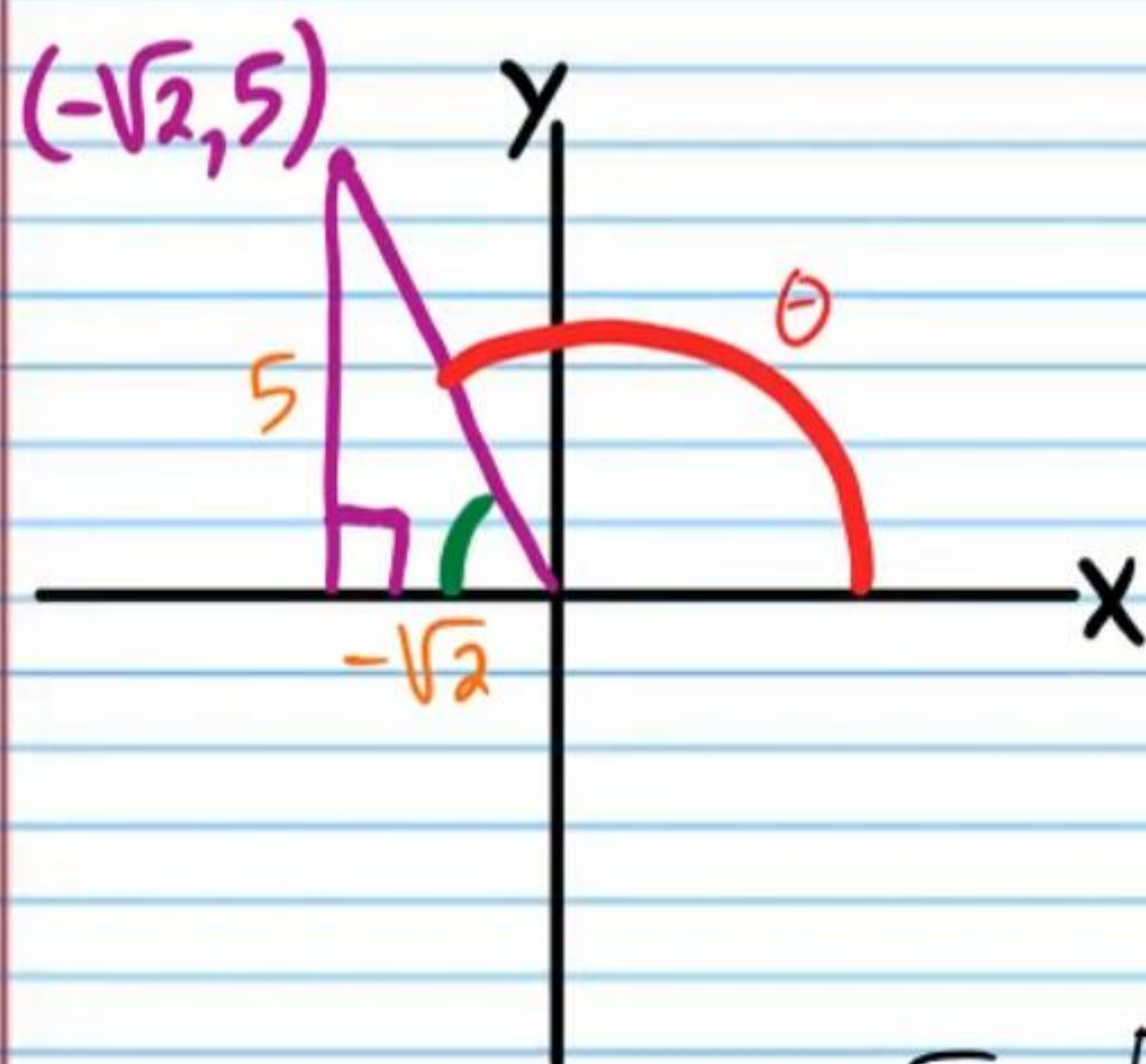


$$\begin{aligned} (-\sqrt{6})^2 + (-\sqrt{7})^2 &= c^2 \\ 6 + 7 &= c^2 \\ 13 &= c^2 \\ c &= \pm \sqrt{13} \\ c &= \sqrt{13} \end{aligned}$$

$$\begin{aligned} \text{i) } \sin \theta &= \frac{-\sqrt{7}}{\sqrt{13}} = \boxed{-\frac{\sqrt{91}}{13}} \\ \text{ii) } \cos \theta &= \frac{-\sqrt{6}}{\sqrt{13}} = \boxed{-\frac{\sqrt{78}}{13}} \\ \text{iii) } \tan \theta &= \frac{-\sqrt{7}}{-\sqrt{6}} = \boxed{\frac{\sqrt{42}}{6}} \\ \text{iv) } \csc \theta &= \frac{\sqrt{13}}{-\sqrt{7}} = \boxed{-\frac{\sqrt{91}}{7}} \\ \text{v) } \sec \theta &= \frac{\sqrt{13}}{-\sqrt{6}} = \boxed{-\frac{\sqrt{78}}{6}} \\ \text{vi) } \cot \theta &= \frac{-\sqrt{6}}{-\sqrt{7}} = \boxed{\frac{\sqrt{42}}{7}} \end{aligned}$$



⑭ $(-\sqrt{2}, 5)$



$$5^2 + (-\sqrt{2})^2 = r^2$$

$$25 + 2 = r^2$$

$$27 = r^2$$

$$r = \pm\sqrt{27}$$

$$r = \sqrt{9 \cdot 3}$$

$$r = \sqrt{9} \cdot \sqrt{3}$$

$$r = 3\sqrt{3}$$

i) $\sin \theta = \frac{y}{r} = \frac{5}{3\sqrt{3}} = \frac{5\sqrt{3}}{3 \cdot 3} = \frac{5\sqrt{3}}{9}$

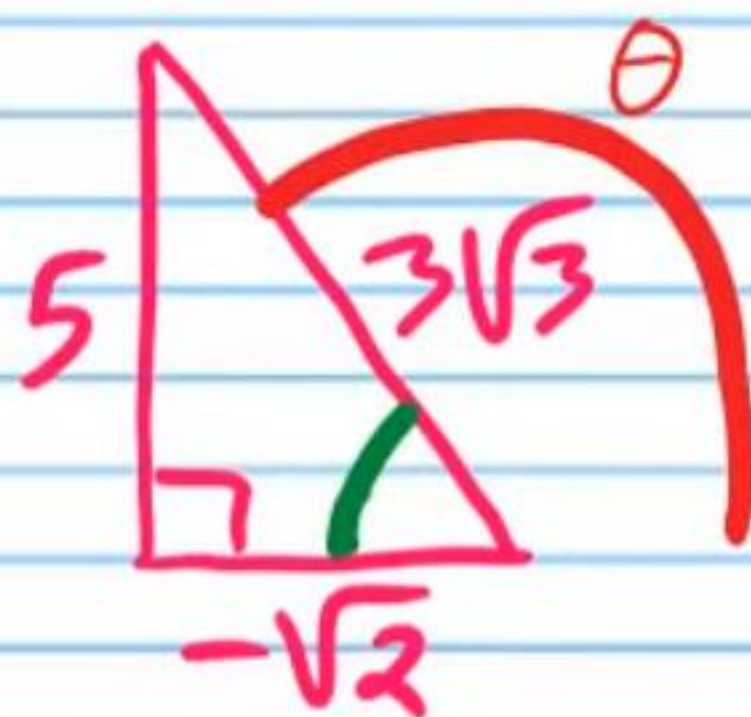
ii) $\cos \theta = \frac{x}{r} = \frac{-\sqrt{2}}{3\sqrt{3}} = -\frac{\sqrt{6}}{3 \cdot 3} = -\frac{\sqrt{6}}{9}$

iii) $\tan \theta = \frac{y}{x} = \frac{5}{-\sqrt{2}} = -\frac{5\sqrt{2}}{2}$

iv) $\csc \theta = \frac{r}{y} = \frac{3\sqrt{3}}{5}$

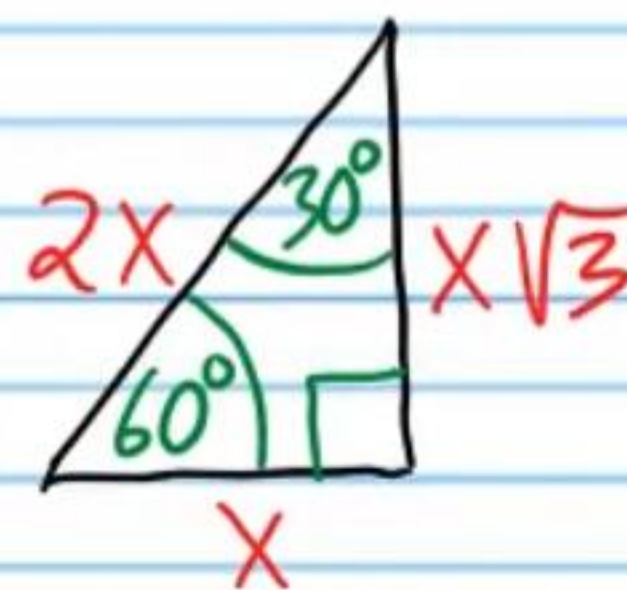
v) $\sec \theta = \frac{r}{x} = \frac{3\sqrt{3}}{-\sqrt{2}} = \frac{3\sqrt{6}}{-2}$

vi) $\cot \theta = \frac{x}{y} = -\frac{\sqrt{2}}{5}$



Special Right Triangles

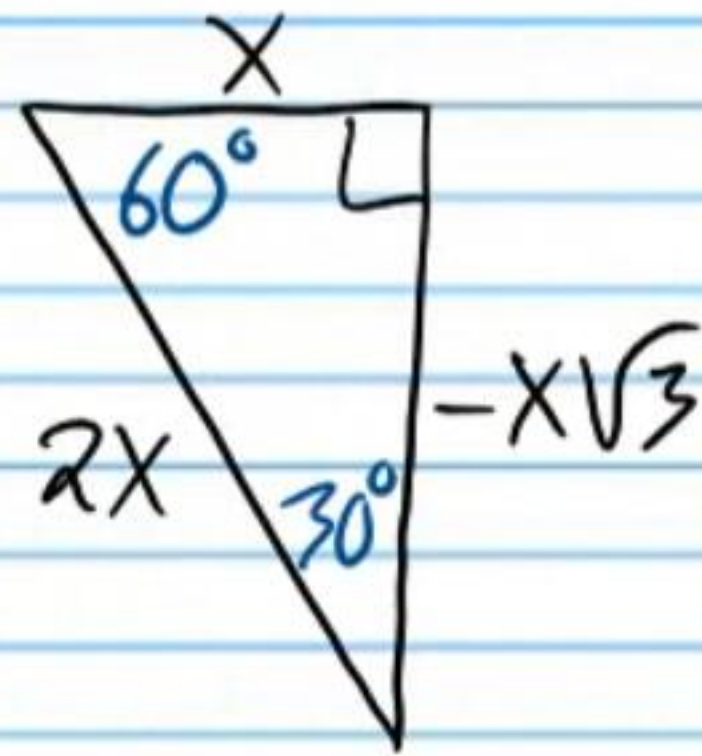
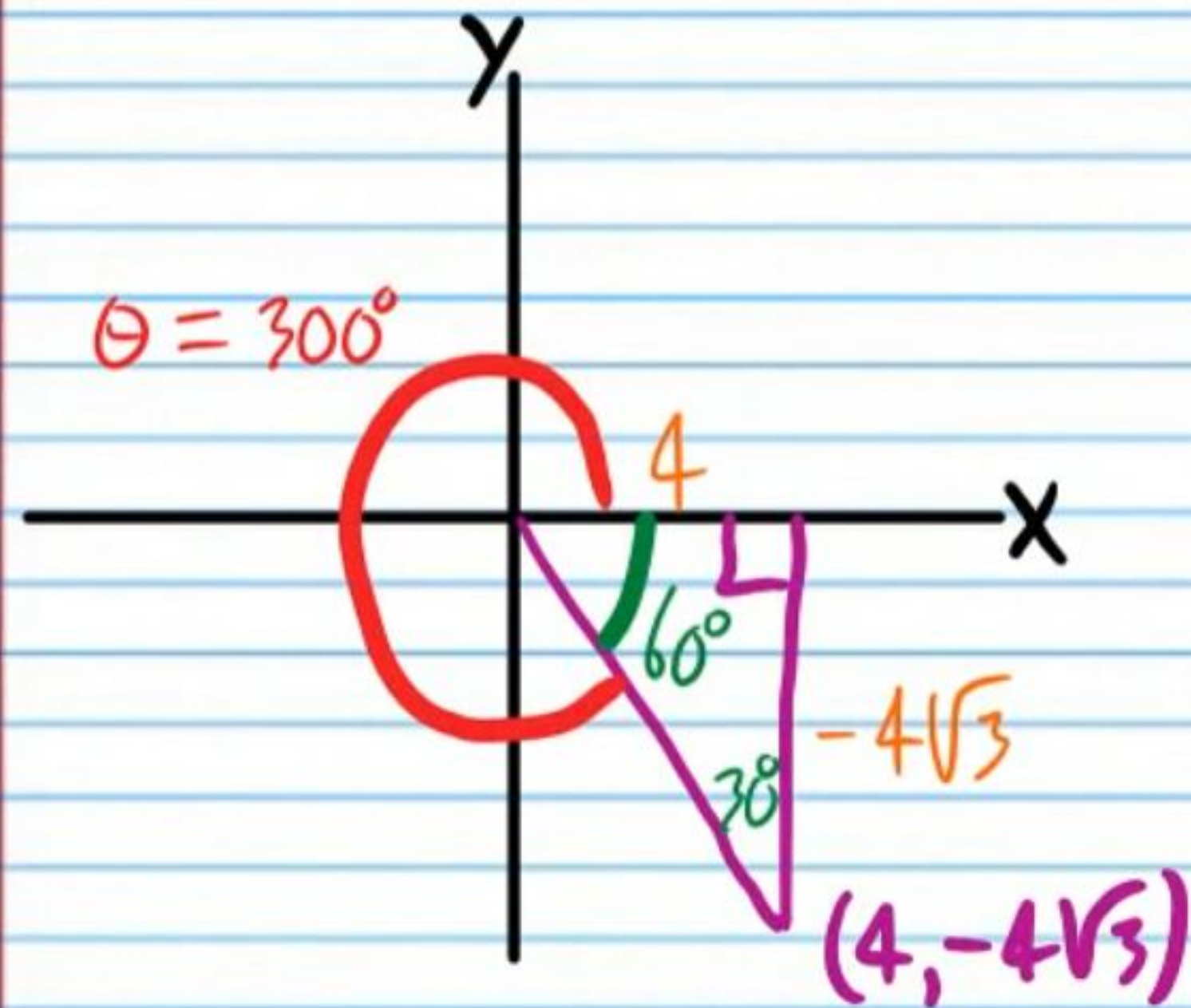
30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is x , then the length of the hypotenuse is $2x$ and the length of the side opposite the 60° angle is $x\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are x , then the length of the hypotenuse is $x\sqrt{2}$.



• ⑮ $(4, -4\sqrt{3})$



$$\text{i) } \sin \theta = \frac{-4\sqrt{3}}{8} = \boxed{-\frac{\sqrt{3}}{2}}$$

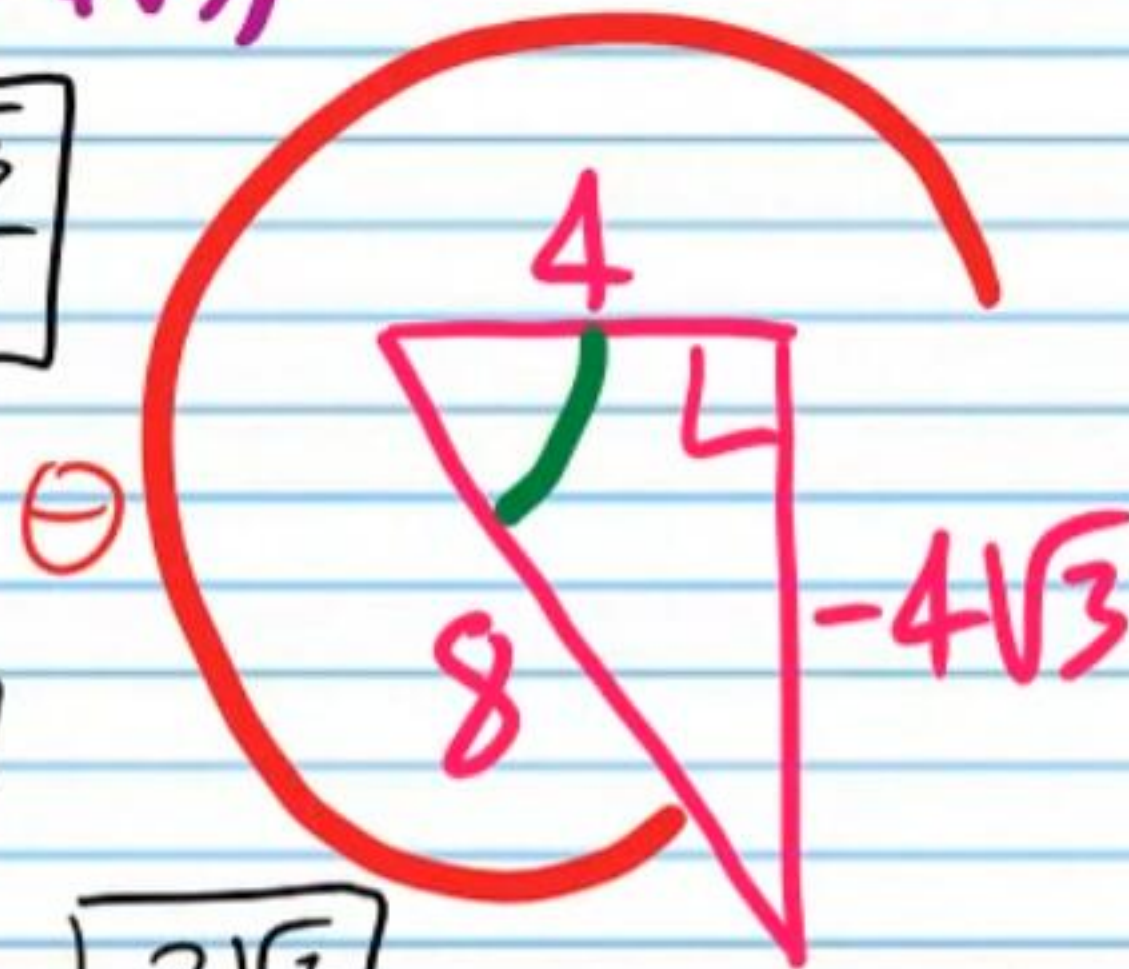
$$\text{ii) } \cos \theta = \frac{4}{8} = \boxed{\frac{1}{2}}$$

$$\text{iii) } \tan \theta = \frac{-4\sqrt{3}}{4} = \boxed{-\sqrt{3}}$$

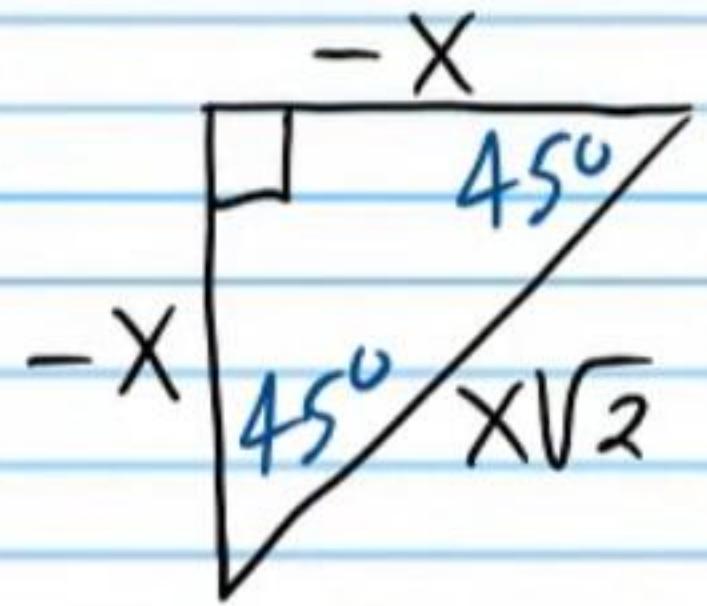
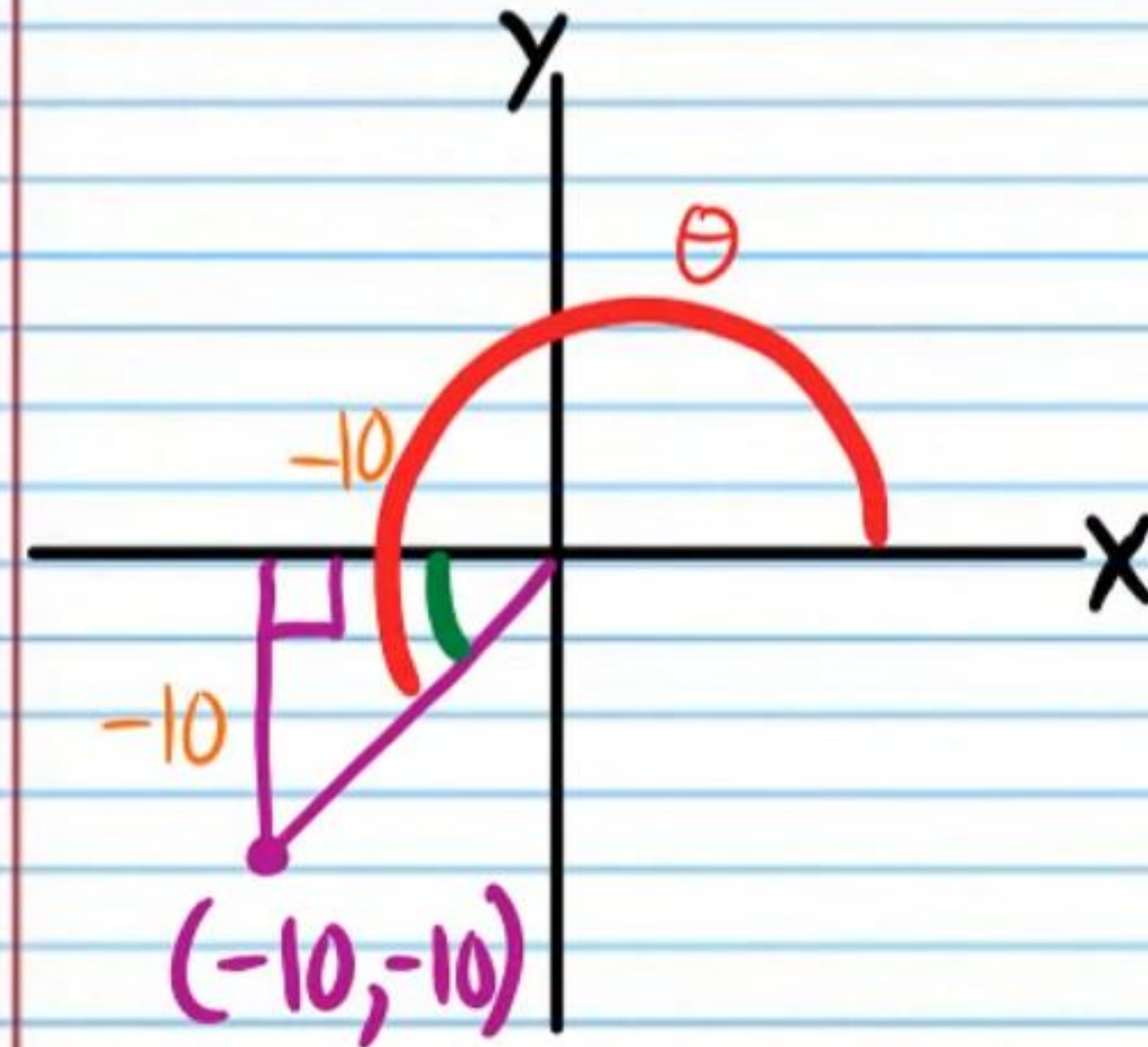
$$\text{iv) } \csc \theta = \frac{8}{-4\sqrt{3}} = -\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$$

$$\text{v) } \sec \theta = \frac{8}{4} = \boxed{2}$$

$$\text{vi) } \cot \theta = \frac{4}{-4\sqrt{3}} = -\frac{1}{\sqrt{3}} = \boxed{-\frac{\sqrt{3}}{3}}$$



⑯ $(-10, -10)$



$$\text{i) } \sin \theta = \frac{-10}{10\sqrt{2}} = -\frac{1}{\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}$$

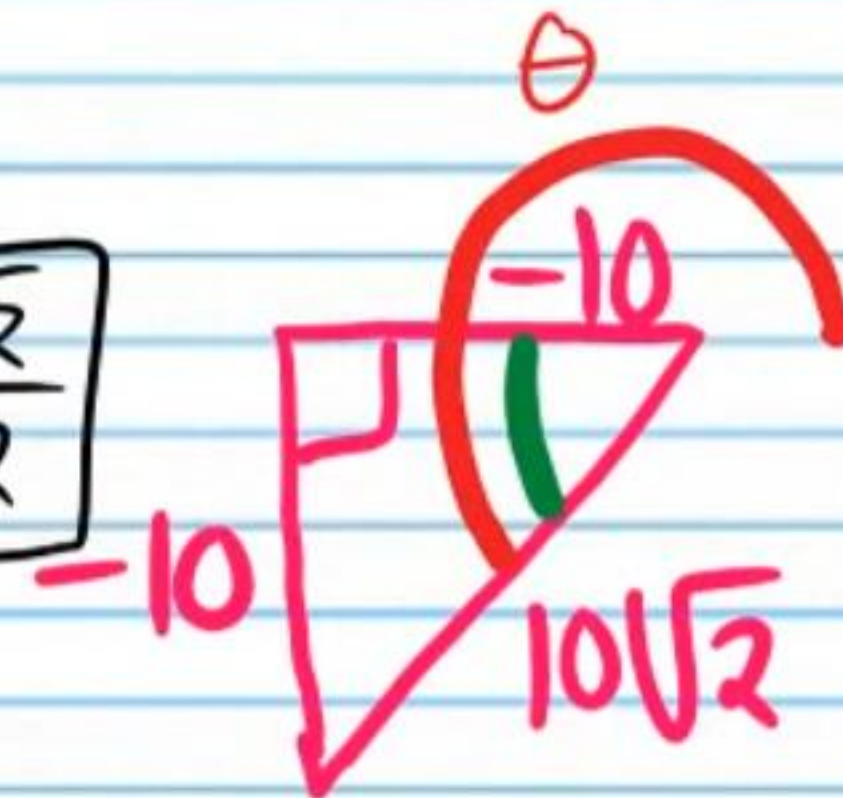
$$\text{ii) } \cos \theta = \frac{-10}{10\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}$$

$$\text{iii) } \tan \theta = \frac{-10}{-10} = \boxed{1}$$

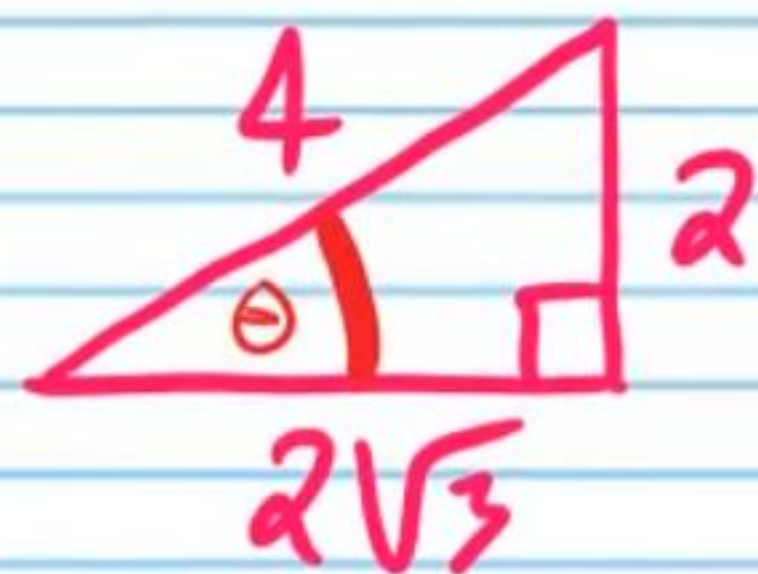
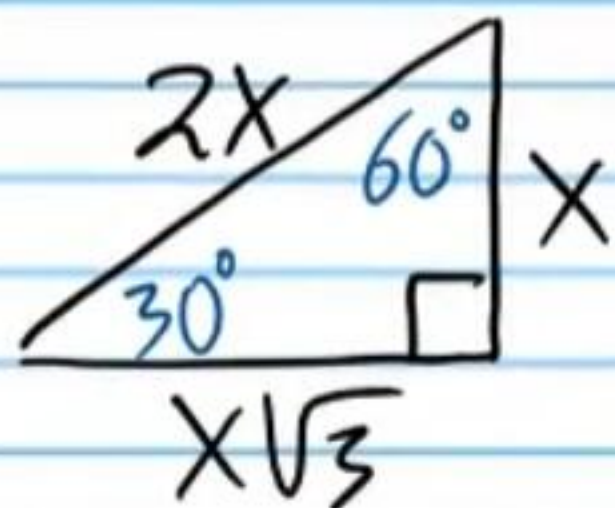
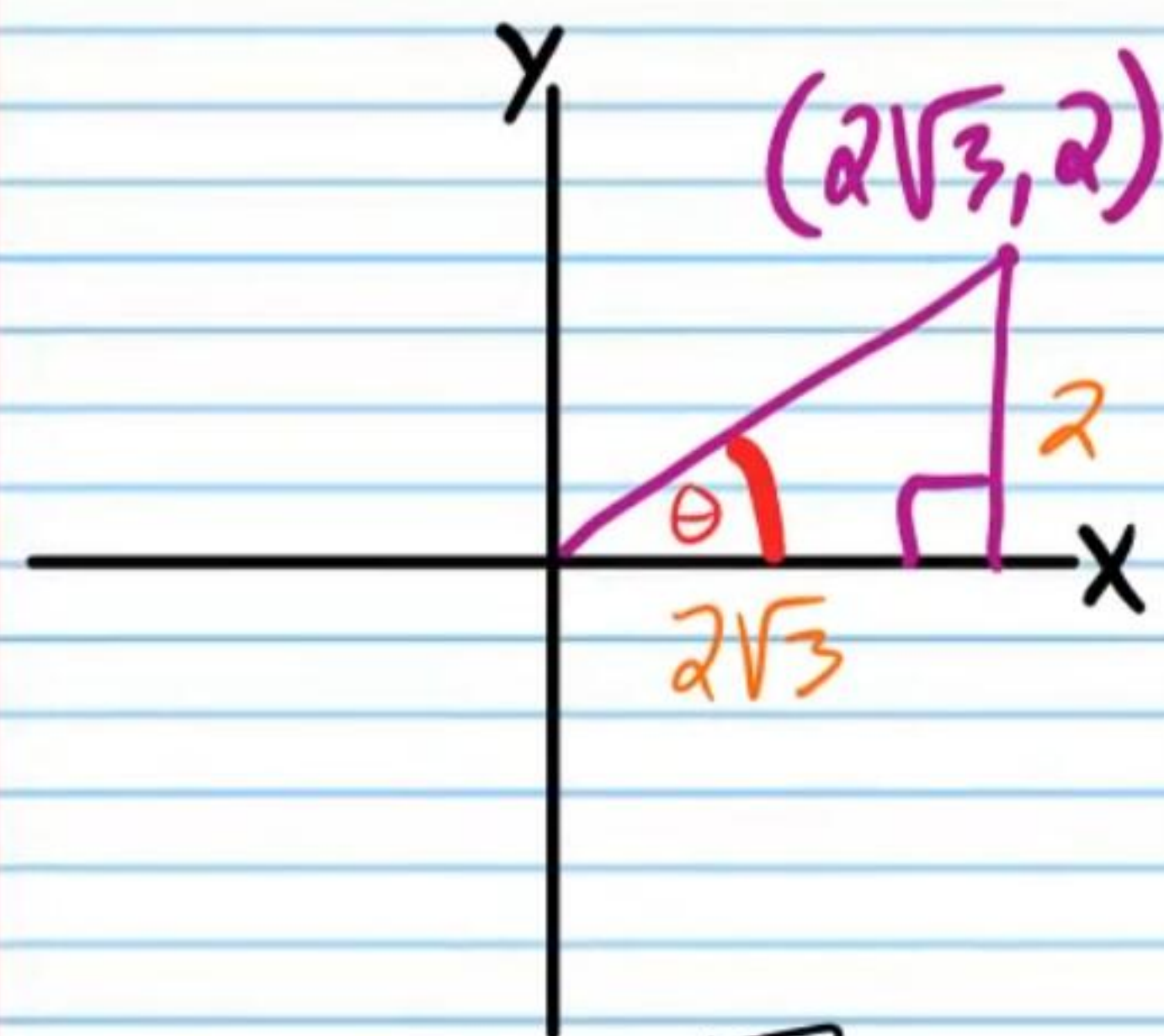
$$\text{iv) } \csc \theta = \frac{10\sqrt{2}}{-10} = \boxed{-\sqrt{2}}$$

$$\text{v) } \sec \theta = \frac{10\sqrt{2}}{-10} = \boxed{-\sqrt{2}}$$

$$\text{vi) } \cot \theta = \frac{-10}{-10} = \boxed{1}$$



• ⑪ $(2\sqrt{3}, 2)$



$$\text{i) } \sin \theta = \frac{2}{4} = \boxed{\frac{1}{2}}$$

$$\text{ii) } \cos \theta = \frac{2\sqrt{3}}{4} = \boxed{\frac{\sqrt{3}}{2}}$$

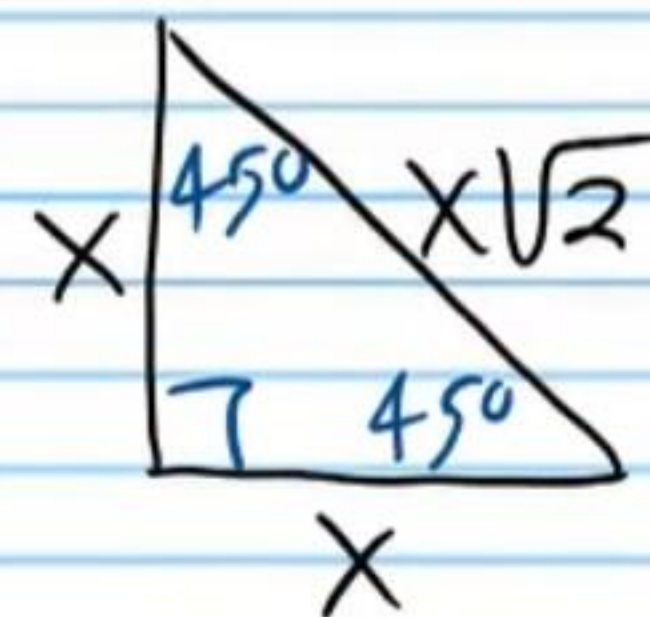
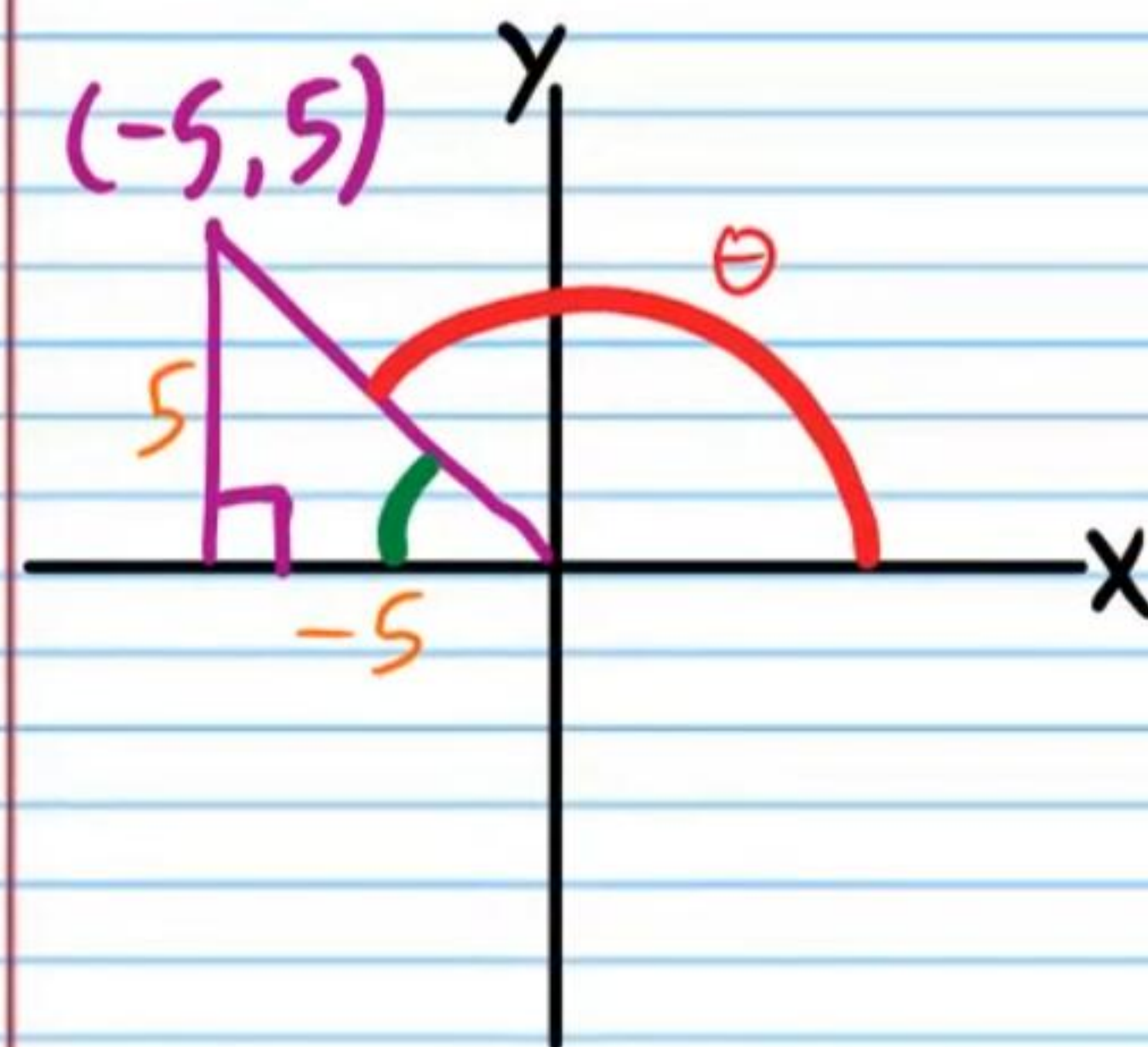
$$\text{iii) } \tan \theta = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$$

$$\text{iv) } \csc \theta = \frac{4}{2} = \boxed{2}$$

$$\text{v) } \sec \theta = \frac{4}{2\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$$

$$\text{vi) } \cot \theta = \frac{2\sqrt{3}}{2} = \boxed{\sqrt{3}}$$

⑫ $(-5, 5)$



$$\text{i) } \sin \theta = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$$

$$\text{ii) } \cos \theta = \boxed{-\frac{\sqrt{2}}{2}}$$

$$\text{iii) } \tan \theta = \frac{5}{-5} = \boxed{-1}$$

$$\text{iv) } \csc \theta = \frac{5\sqrt{2}}{5} = \boxed{\sqrt{2}}$$

$$\text{v) } \sec \theta = \boxed{-\sqrt{2}}$$

$$\text{vi) } \cot \theta = \boxed{-1}$$

